



# **Department of Computer Science**

## **Lab Manual**

**of**

## **MACHINE LEARNING**

**MCA521-4**

**Class Name: IV MCA**

**Master of Computer Application**

**2026**

**Prepared by: Sharon Mathew**

**Verified by:**

|                                                                                     |
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## Department Overview

Department of Computer Science of CHRIST (Deemed to be University) strives to shape outstanding computer professionals with ethical and human values to reshape nation's destiny. The training imparted aims to prepare young minds for the challenging opportunities in the IT industry with a global awareness rooted in the Indian soil, nourished and supported by experts in the field.

## Vision

The Department of Computer Science endeavours to imbibe the vision of the University “**Excellence and Service**”. The department is committed to this philosophy which pervades every aspect and functioning of the department.

## Mission

“To develop IT professionals with ethical and human values”. To accomplish our mission, the department encourages students to apply their acquired knowledge and skills towards professional achievements in their career. The department also moulds the students to be socially responsible and ethically sound.

## Introduction to the Programme

Master of Computer Applications (MCA) is a post-graduate programme of CHRIST (Deemed to be University) and approved by the All India Council of Technical Education (AICTE). It is a two-year programme spread over six trimesters to bridge the chasm between computer studies and applications, bringing within reach of enthusiastic youngsters an excellent group of experienced and dedicated staff and experts, and an enviable infrastructure. MCA at CHRIST (Deemed to be University) strives to shape outstanding computer professionals with ethical and human values to reshape the nation's destiny. The training programme aims to prepare young minds for challenging opportunities in the IT industry with a global awareness rooted in the Indian soil, nourished and supported by experts in the field. The Department is committed to the motto “Excellence and Service,” and this philosophy pervades every aspect of its functioning.

## Programme Objective

This programme is conceptualised and designed based on the strong commitment of the department to provide better quality education to the students. The principal objectives of this course are to:

- Heighten technological awareness
- Train future industry specialists
- Encourage effective software development
- Provide research training
- Involve students in innovative research
- Produce graduates with a two-year professional education in Computer Science with technical, professional, and communications skills.

## Ethics and Human Values

1. Only proprietary or open-source software would be used for academic teaching and learning purposes.
2. Copying of programs from the internet, friends or from other sources is strictly discouraged since it impairs development of programming skills.
3. Unique Practical (Domain based) exercises ensure that the students don't get involved in code plagiarism.
4. Projects undertaken by students during the course are done in teams to improve collaborative work and synergy between team members.
5. Projects involve modularization which initiates students to take individual responsibility for common goals.
6. Passion for excellence is promoted among the students, be it in software development or project documentation.
7. Giving due credit to sources during the seminar and research assignment is promoted among the students
8. The course and its design enforce the practice of good referencing technique to improve the sense of integrity.
9. Courses involving group discussions and debates on ethical practices and human values are designed to sensitize the students in dealing with customers and members within the Organization.

### **Programme Outcomes:**

- P01: Foundation Knowledge: Apply knowledge of mathematics, programming logic, and coding fundamentals for solution architecture and problem-solving.
- P02: Problem Analysis: Identify, review, formulate, and analyse problems primarily focussing on customer requirements using critical thinking frameworks.
- P03: Development of Solutions: Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.
- P04: Modern Tool Usage: Select, adapt, and apply modern computational tools such as the development of algorithms with an understanding of the limitations including human biases.
- P05: Individual and Teamwork: Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.
- P06: Project Management and Finance: Use the principles of project management such as scheduling, and work breakdown structure and be conversant with the principles of Finance for profitable project management.
- P07: Ethics: Commit to professional ethics in managing software projects with financial aspects.
- Learn to use new technologies for cyber security and insulate customers from malware
- P08: Life-long learning: Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

### **Programme Educational Objectives(PEO's)**

- PEO1: Develop innovative and effective software solutions through research that addresses real-world challenges, grounded in a commitment to continuous learning and adaptability.
- PEO2: Advance the knowledge and practices in the field of computer applications through lifelong learning and rigorous research, supporting professional growth and academic excellence.
- PEO3: Exhibit ethical leadership, social responsibility to contribute and enhance community well-being by innovating solutions.

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## **MCA521-4 – MACHINE LEARNING**

**Course Name: MACHINE LEARNING Code: MCA521-4**

**Total Hours: 75**

**L + P + T: 3 + 4 + 0**

**Max Marks: 100**

**Credits: 4**

**Course Type: Core Course**

**Course Category: Theo&Prac**

### **Course Description**

Machine Learning is a key area in computer applications that focuses on building models capable of making predictions or informed decisions based on data. A strong understanding of machine learning enables students to analyze and interpret complex datasets, develop predictive models, and extract meaningful insights. This course covers a wide range of machine learning concepts and techniques, including supervised and unsupervised learning. It provides a solid theoretical foundation while emphasizing practical, hands-on experience with algorithms and real-world data. The course equips students with the knowledge and skills required to apply machine learning techniques across various domains using modern tools and methodologies.

### **Course Objectives**

This course enables students to understand the fundamental concepts of Machine Learning, including its core principles, terminology, and methodologies. Students will learn to differentiate between various types of learning algorithms such as supervised, unsupervised, and reinforcement learning, and recognize their appropriate applications.

### **Course Outcomes**

After successful completion of the course, students will be able to:

- Explain the fundamental principles and scope of Machine Learning
- Implement modelling practices in machine learning for better generalisation
- Interpret predictive modelling results and performance of supervised machine learning techniques
- Assess the outcomes and effects of unsupervised machine learning processes
- Deploy robust machine learning models for interdisciplinary applications

### **Unit-1: INTRODUCTION TO MACHINE LEARNING Teaching Hours: 15 Hours**

Introduction to AI – Knowledge Representation – Data – Representation of Data, Data Processing, Data Storage, Data Processing, Types of Data – Exploratory Data Analysis, Visualisations and Interpretation, Outliers

Machine Learning: Definition, Objectives, Components, Features, Applications – Traditional Programming v/s Machine Learning, Types of Machine Learning – Predictive Models, Statistical Inferences – Applications of Machine Learning

Challenges in ML: Insufficient Quantity of Data, Non-representative and Poor-Quality Data, Irrelevant Features, Estimation Estimation, Reducing Human Biases in Model Building, Ethical Use of Technology

Lab Exercises:

1. Data Preprocessing and visualisation
2. Data exploration and inferences

**Unit-2: PROCESSES IN MACHINE LEARNING Teaching Hours: 15 Hours**

Data Preparation and Model Selection: Effect of Data Preparation: Encoding, Nominal and Ordinal Representation, Feature Transformation and Feature Engineering, Generalisation, Normalisation, Sampling, Using Saved Weights, Model Selection Strategies: Overfitting, Underfitting, Bias-Variance Trade-off, Testing and Validation: Performance measures - Confusion matrix, Precision-Recall Trade off, F1 Score, ROC Curve, AUC, Error Analysis, Model Parameters, Hyperparameters, Hyperparameter Tuning, Cross Validation, Decision Functions: Introduction to Supervised Machine Learning Methods, Decision Functions, Decision Thresholds, Distance Measures, Hyperplanes, Regression & Classification Problems, Loss and Cost Functions, Linear Regression using OLS, Multi-class vs Multi-label, KNN Classification

Lab Exercises:

3. Saving weights of a generalised model.
4. Regression and Classification Evaluation Metrics: Illustration through Linear Regression and KNN Classification

**Unit-3: SUPERVISED LEARNING ALGORITHMS****Teaching Hours: 15****Hours**

Learning through Optimisation: Gradient Descent, Linear Regression, Logistic Regression  
Computational Learning: Decision Trees, Naive-bayes classification, Support Vector Machines, Ensemble Learners: Learning, Voting, Bagging, Stacking, Boosting, Random Forests, GridSearchCV

Lab Exercises:

5. Regression through Gradient Descent
6. Comparison of Different Classifiers
7. Illustration of Ensemble Learning Methods

**Unit-4: UNSUPERVISED LEARNING AND DIMENSIONALITY REDUCTION Teaching Hours: 15 Hours**

Introduction: Types of Unsupervised Learning, Challenges in Unsupervised Learning, Concept of Cost Functions in Unsupervised Learning, Clustering Methods: Distance based, Centroid Based, Elbow method to find Optimal Number of Clusters, Silhouette Method, K-Means Clustering, Hierarchical Clustering: Agglomerative and Bottom Up, Applying Supervised Learning after Clustering, Dimensionality Reduction Methods: Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA)

Lab Exercises:

8. KMeans and Elbow Methods
9. Hierarchical Clustering and Dendrograms
10. Dimensionality Reduction
11. Image Compression using Dimensionality Reduction.

**Unit-5 Teaching Hours: 15 Hours****MACHINE LEARNING IN REAL-WORLD**

Emerging Techniques: Role of ML in attaining Sustainable Development Goals, Time Series Preprocessing, Blackbox Methods: Neural Networks & Deep Learning, Reinforcement Learning/QLearning, Self Supervised Learning, Quantum Machine Learning, Keras and Tensorflow for designing Multi Layer Perceptron. Case Studies: CNN, RNN, Advances in Deep Learning, Natural Language Processing, Real Time Systems, Computer Vision, Robotics, Expert Systems, Generative Networks, Large Language Models. Model Deployment: Model Deployment: Connecting Models to User Interface, Deployment of ML Models.

Lab Exercises:

12. Training a Multi-layer perceptron

13. Deploying Trained ML-models

**Text and Reference Books**

[1] Campesato, O. (2020). Artificial Intelligence, Machine Learning and Deep Learning. Mercury Learning and Information.

[2] Andreas C. Müller and Sarah Guido (2017), "Introduction to Machine Learning with Python A Guide For Data Scientists" O'Reilly book

[3] Ethem Alpaydin (2005), "Introduction to Machine Learning", Prentice Hall of India

[4] Max Kuhn and Kjell Johnson (2018), "Applied Predictive Modelling", Springer

**Essential Reading**

[1] Stuart Russell and Peter Norvig(2020), "Artificial Intelligence: A Modern Approach" (4th Edition), Pearson

[2] Aurelien Geron(2019), "Hands on Machine Learning with Scikit-learn, Keras and Tensorflow", 2nd Edition, O'Reilly Books

[3] Kevin P. Murphy(2012), "Machine Learning: A Probabilistic Perspective", MIT Press

[4] Hastie, Tibshirani, Friedman(2008), "The Elements of Statistical Learning" (2nd ed), Springer

[5] Ian Goodfellow, Aaron Courville, Yoshua Bengio (2019), "Deep Learning (Adaptive Computation And Machine Learning Series)", MIT Press

Additional Information (Web Resources):

1. Andrew Ng, Deeplearning.ai, Machine Learning Specialisation, offered through Coursera:

<https://www.deeplearning.ai/courses/machine-learning-specialization/>

Evaluation Pattern: CIA - 50% ESE - 50%

## LIST OF PROGRAMS

### MCA-B

| Sl. no | Title of lab Experiment                                                                                     | Page number | RB T | CO     |
|--------|-------------------------------------------------------------------------------------------------------------|-------------|------|--------|
| 1      | India Air Quality & Crop Yield — EDA Lab Data Preprocessing, Visualisation & Exploration Combined Lab Sheet | 8           | L3   | CO1, 3 |
| 2      | India Air Quality & Crop Yield — EDA Lab Data Preprocessing, Visualisation & Exploration Combined Lab Sheet | 8           | L3   | CO2, 3 |
| 3      | Student Survey: Simple Linear Regression using Scikit Learn, OLS, and Parameter Saving                      | 46          | L3   | CO3    |
| 4      | Regression and Classification Evaluation Metrics                                                            | 76          | L3   | CO3    |
| 5      |                                                                                                             |             | L3   | CO3    |
| 6      |                                                                                                             |             | L3   | CO3, 4 |
| 7      |                                                                                                             |             | L4   | CO3    |
| 8      |                                                                                                             |             | L3   | CO3    |
| 9      |                                                                                                             |             | L3   | CO4    |
| 10     |                                                                                                             |             | L5   | CO4    |

### Evaluation Rubrics:

Evaluation Rubrics for each program would be

### Submission Guidelines:

- Make a copy of the lab manual template with your <name\_reg:no\_subject name > ,
- Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as lab number.

- Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
- Create a Git Repository in your profile <ML lab-reg no> . Follow a different branch for each lab <Lab 1, Lab 2...>, and push the code to Git. The link should be provided in Google Classroom along with the PDF of the lab manual.
- Upload the PDF to Google Classroom before the deadline.

## Lab 1

### Lab Exercise I:

#### Aim

- To investigate whether worsening air quality across Indian states is linked to declining agricultural output by independently choosing, applying, and justifying appropriate data analysis and visualisation techniques on raw, messy, real-world datasets.
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### Evaluation Rubrics

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### Submission Guidelines

1. Make a copy of the lab manual template with your <name\_reg:no\_subject name>
2. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
3. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
4. Create a **Git repository** in your profile <ML lab-reg no>. Follow a different branch for each lab <Lab 1, Lab 2 ...> and push the code to Git.
5. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
6. Upload the PDF to Google Classroom before the deadline.



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## Code with Results

### TASK 1:

A junior analyst hands you two CSV files and says — "I have no idea what's in these. We need to use them to build a machine learning model next week." Before any analysis can begin, you need to understand what you are working with.

Investigate both files thoroughly. Produce a structured summary that gives a complete first-impression picture of the data — its size, its contents, and where it might be problematic. Decide what information a data scientist would need before trusting this data, and make sure your summary covers all of it.

A structured data profile for each file in a markdown cell

At least one written observation about what concerns you after the inspection

*Think: What does a data scientist need to know about a dataset before using it? What functions help you find that?*

### CODE:

```
import pandas as pd
import numpy as np

city_df = pd.read_csv("city_day.csv")
crop_df = pd.read_csv("crop_production.csv")

def data_profile(df, dataset_name):
    print("\n" + "="*80)
    print(f"DATA PROFILE : {dataset_name}")
    print("="*80)

    print("\n1. Dataset Shape")
    print(f"Rows      : {df.shape[0]}")
    print(f"Columns   : {df.shape[1]}")
```

```
print("\n2. Data Types")
print(df.dtypes)

print("\n3. First 5 Records")
print(df.head())

print("\n4. Missing Values")
missing = pd.DataFrame({
    'Missing_Count': df.isnull().sum(),
    'Missing_%': round((df.isnull().sum()/len(df))*100,2)
}).sort_values(by='Missing_%', ascending=False)

print(missing[missing['Missing_Count'] > 0])

print("\n6. Duplicate Rows")
print(df.duplicated().sum())

print("\n7. Numerical Summary")
print(df.describe())

print("\n8. Categorical Summary")
print(df.describe(include='object'))

print("\n9. Unique Values")
print(df.nunique())

# Run Profiles
data_profile(city_df, "City Air Quality Dataset")
data_profile(crop_df, "Crop Production Dataset")
```

## RESULT

```
=====
DATA PROFILE : City Air Quality Dataset
=====
```

### 1. Dataset Shape

```
Rows      : 29531
Columns   : 16
```

### 2. Data Types

```
City      object
Date      object
PM2.5     float64
PM10      float64
NO        float64
NO2       float64
NOx       float64
NH3       float64
CO        float64
SO2       float64
O3        float64
Benzene   float64
Toluene   float64
Xylene    float64
AQI       float64
AQI_Bucket object
dtype: object
```

### 3. First 5 Records

|   | City      | Date       | PM2.5 | PM10 | NO    | NO2   | NOx   | NH3 | CO    | SO2   | \ |
|---|-----------|------------|-------|------|-------|-------|-------|-----|-------|-------|---|
| 0 | Ahmedabad | 2015-01-01 | NaN   | NaN  | 0.92  | 18.22 | 17.15 | NaN | 0.92  | 27.64 |   |
| 1 | Ahmedabad | 2015-01-02 | NaN   | NaN  | 0.97  | 15.69 | 16.46 | NaN | 0.97  | 24.55 |   |
| 2 | Ahmedabad | 2015-01-03 | NaN   | NaN  | 17.40 | 19.30 | 29.70 | NaN | 17.40 | 29.07 |   |
| 3 | Ahmedabad | 2015-01-04 | NaN   | NaN  | 1.70  | 18.48 | 17.97 | NaN | 1.70  | 18.59 |   |
| 4 | Ahmedabad | 2015-01-05 | NaN   | NaN  | 22.10 | 21.42 | 37.76 | NaN | 22.10 | 39.33 |   |

|   | O3     | Benzene | Toluene | Xylene | AQI | AQI_Bucket |
|---|--------|---------|---------|--------|-----|------------|
| 0 | 133.36 | 0.00    | 0.02    | 0.00   | NaN | NaN        |
| 1 | 34.06  | 3.68    | 5.50    | 3.77   | NaN | NaN        |
| 2 | 30.70  | 6.80    | 16.40   | 2.25   | NaN | NaN        |
| 3 | 36.08  | 4.43    | 10.14   | 1.00   | NaN | NaN        |
| 4 | 39.31  | 7.01    | 18.89   | 2.78   | NaN | NaN        |

---

#### 4. Missing Values

|            | Missing_Count | Missing_% |
|------------|---------------|-----------|
| Xylene     | 18109         | 61.32     |
| PM10       | 11140         | 37.72     |
| NH3        | 10328         | 34.97     |
| Toluene    | 8041          | 27.23     |
| Benzene    | 5623          | 19.04     |
| AQI        | 4681          | 15.85     |
| AQI_Bucket | 4681          | 15.85     |
| PM2.5      | 4598          | 15.57     |
| NOx        | 4185          | 14.17     |
| O3         | 4022          | 13.62     |
| SO2        | 3854          | 13.05     |
| NO2        | 3585          | 12.14     |
| NO         | 3582          | 12.13     |
| CO         | 2059          | 6.97      |

#### 6. Duplicate Rows

0

## 8. Categorical Summary

|        | City      | Date       | AQI_Bucket |
|--------|-----------|------------|------------|
| count  | 29531     | 29531      | 24850      |
| unique | 26        | 2009       | 6          |
| top    | Ahmedabad | 2020-06-26 | Moderate   |
| freq   | 2009      | 26         | 8829       |

## 9. Unique Values

|            |       |
|------------|-------|
| City       | 26    |
| Date       | 2009  |
| PM2.5      | 11716 |
| PM10       | 12571 |
| NO         | 5776  |
| NO2        | 7404  |
| NOx        | 8156  |
| NH3        | 5922  |
| CO         | 1779  |
| SO2        | 4761  |
| O3         | 7699  |
| Benzene    | 1873  |
| Toluene    | 3608  |
| Xylene     | 1561  |
| AQI        | 829   |
| AQI_Bucket | 6     |

dtype: int64

## 7. Numerical Summary

|       | Crop_Year     | Area         | Production   |
|-------|---------------|--------------|--------------|
| count | 246091.000000 | 2.460910e+05 | 2.423610e+05 |
| mean  | 2005.643018   | 1.200282e+04 | 5.825034e+05 |
| std   | 4.952164      | 5.052340e+04 | 1.706581e+07 |
| min   | 1997.000000   | 4.000000e-02 | 0.000000e+00 |
| 25%   | 2002.000000   | 8.000000e+01 | 8.800000e+01 |
| 50%   | 2006.000000   | 5.820000e+02 | 7.290000e+02 |
| 75%   | 2010.000000   | 4.392000e+03 | 7.023000e+03 |
| max   | 2015.000000   | 8.580100e+06 | 1.250800e+09 |

## 8. Categorical Summary

|        | State_Name    | District_Name | Season | Crop   |
|--------|---------------|---------------|--------|--------|
| count  | 246091        | 246091        | 246091 | 246091 |
| unique | 33            | 646           | 6      | 124    |
| top    | Uttar Pradesh | BIJAPUR       | Kharif | Rice   |
| freq   | 33306         | 945           | 95951  | 15104  |

## TASK 2:

After the inspection, it is clear that several columns have missing values. A colleague suggests simply deleting all rows with any null value. Another suggests filling everything with the mean. You are not sure either approach is right.

Devise your own missing value treatment strategy. For each column with missing data, decide whether to drop it, drop affected rows, or impute — and explain why that choice is appropriate for that specific column. Apply your strategy and verify it worked.

A markdown cell that justifies each decision column by column  
Before and after null counts as evidence the treatment worked.

```
city_df.drop(columns=['Xylene'], inplace=True)

num_cols = [
    'PM2.5', 'PM10', 'NO', 'NO2', 'NOx',
    'NH3', 'CO', 'SO2', 'O3',
    'Benzene', 'Toluene', 'AQI'
]

for col in num_cols:
    city_df[col].fillna(city_df[col].median(), inplace=True)

city_df['AQI_Bucket'].fillna(
    city_df['AQI_Bucket'].mode()[0],
    inplace=True
)

print("Missing Values After Treatment")
print(city_df.isnull().sum())

print("\nTotal Missing Values:",
```

---

```
city_df.isnull().sum().sum()
```

**RESULT:**

```
Missing Values After Treatment
City          0
Date          0
PM2.5         0
PM10          0
NO            0
NO2           0
NOx           0
NH3           0
CO            0
SO2           0
O3            0
Benzene       0
Toluene       0
AQI           0
AQI_Bucket    0
dtype: int64
```

---

**TASK 3:**

You need to combine both files later in the lab using the State column as the common key. But when you look closely, the same state appears with different spellings, extra spaces, or old names across the two files. There are also rows that appear more than once for no clear reason.

Make both files agree on state names and remove any redundant records. Document every inconsistency you found and how you resolved it. Show that the files are now ready to be merged reliably.

A list of all inconsistencies found and the fix applied for each  
Record counts before and after to prove duplicates were removed

```
print("Before Treatment")
print(crop_df.isnull().sum())

# -----
# Remove rows with missing Production
# -----

crop_df = crop_df.dropna(subset=['Production'])

# -----
# Verification
# -----

print("\nAfter Treatment")
print(crop_df.isnull().sum())

print("\nTotal Missing Values:",
      crop_df.isnull().sum().sum())
```



**RESULT:**

```
Before Treatment
State_Name      0
District_Name   0
Crop_Year       0
Season          0
Crop            0
Area            0
Production      3730
dtype: int64
```

```
After Treatment
State_Name      0
District_Name   0
Crop_Year       0
Season          0
Crop            0
Area            0
Production      0
dtype: int64
```

```
Total Missing Values: 0
```

```
print("Unique States:")
print(sorted(crop_df['State_Name'].unique()))
```

**Result:** ['Bihar', 'Chandigarh', 'Chhattisgarh', 'Dadra and Nagar Haveli', 'Goa', 'Gujarat', 'Haryana', 'Himachal Pradesh', 'Jammu and Kashmir ', 'Jharkhand', 'Karnataka', 'Kerala', 'Madhya Pradesh', 'Maharashtra', 'Manipur', 'Meghalaya', 'Mizoram', 'Nagaland', 'Odisha', 'Puducherry', 'Punjab', 'Rajasthan', 'Sikkim', 'Tamil Nadu', 'Telangana ', 'Tripura', 'Uttar Pradesh', 'Uttarakhand', 'West Bengal']

```
def clean_state_name(state):

    if pd.isna(state):

        return state

    state = str(state).strip().upper()
```

```

replacements = {
    'TAMILNADU': 'TAMIL NADU',
    'TAMIL NADU ': 'TAMIL NADU',
    'PONDICHERRY': 'PUDUCHERRY',
    'ORISSA': 'ODISHA',
    'UTTARANCHAL': 'UTTARAKHAND',
    'ANDAMAN AND NICOBAR ISLANDS':
        'ANDAMAN & NICOBAR ISLANDS'
}

return replacements.get(state, state)

crop_df['State_Name'] = crop_df['State_Name'].apply(clean_state_name)

```

```

states_before = sorted(
    pd.read_csv("crop_production.csv")
    ['State_Name']
    .dropna()
    .unique()
)

print(states_before)

states_after = sorted(
    crop_df['State_Name']
    .dropna()
    .unique()
)

```

```
print(states_after)
```

**Result:** ['Andaman and Nicobar Islands', 'Andhra Pradesh', 'Arunachal Pradesh', 'Assam', 'Bihar', 'Chandigarh', 'Chhattisgarh', 'Dadra and Nagar Haveli', 'Goa', 'Gujarat', 'Haryana', 'Himachal Pradesh', 'Jammu and Kashmir ', 'Jharkhand', 'Karnataka', 'Kerala', 'Madhya Pradesh', 'Maharashtra', 'Manipur', 'Meghalaya', 'Mizoram', 'Nagaland', 'Odisha', 'Puducherry', 'Punjab', 'Rajasthan', 'Sikkim', 'Tamil Nadu', 'Telangana ', 'Tripura', 'Uttar Pradesh', 'Uttarakhand', 'West Bengal']

```
['ANDAMAN & NICOBAR ISLANDS', 'ANDHRA PRADESH', 'ARUNACHAL PRADESH',
'ASSAM', 'BIHAR', 'CHANDIGARH', 'CHHATTISGARH', 'DADRA AND NAGAR
HAVELI', 'GOA', 'GUJARAT', 'HARYANA', 'HIMACHAL PRADESH', 'JAMMU AND
KASHMIR', 'JHARKHAND', 'KARNATAKA', 'KERALA', 'MADHYA PRADESH',
'MAHARASHTRA', 'MANIPUR', 'MEGHALAYA', 'MIZORAM', 'NAGALAND', 'ODISHA',
'PUDUCHERRY', 'PUNJAB', 'RAJASTHAN', 'SIKKIM', 'TAMIL NADU',
'TELANGANA', 'TRIPURA', 'UTTAR PRADESH', 'UTTARAKHAND', 'WEST BENGAL']
```

```
before_crop = len(crop_df)

crop_df = crop_df.drop_duplicates()

after_crop = len(crop_df)

print("Crop Dataset")
print("Before:", before_crop)
print("After :", after_crop)
print("Duplicates Removed:",
      before_crop - after_crop)

city_df = pd.read_csv("city_day.csv")

before_city = len(city_df)
```

---

```
city_df = city_df.drop_duplicates()

after_city = len(city_df)

print("City Dataset")
print("Before:", before_city)
print("After :", after_city)
print("Duplicates Removed:",
      before_city - after_city)

print("City duplicates:",
      city_df.duplicated().sum())

print("Crop duplicates:",
      crop_df.duplicated().sum())

city_df.to_csv(
    "city_day_cleaned.csv",
    index=False
)

crop_df.to_csv(
    "crop_production_cleaned.csv",
    index=False
)
```

## Result:

```
Crop Dataset
Before: 246091
After : 246091
Duplicates Removed: 0
City Dataset
Before: 29531
After : 29531
Duplicates Removed: 0
City duplicates: 0
Crop duplicates: 0
```

---

**TASK 4:**

The pollution control board wants to know — are most Indian cities moderately polluted, or is the problem concentrated in just a few places? They also want to know whether the average AQI is a fair number to report publicly, or whether extreme cities are pulling it up unfairly.

Investigate the shape of the AQI distribution. Decide which visualisation(s) best answer both questions the board is asking, justify your choice, produce the plot(s), and record two specific observations that directly address their concerns.

Your chosen plot(s) with fully labelled axes and a descriptive title

A markdown cell with your justification for the chosen plot type and two observations

*Think: Which plot shows where values cluster? Which one reveals extreme values? Do you need one plot or two to answer both questions?*

```
import pandas as pd
import matplotlib.pyplot as plt

# Remove missing AQI values
aqi = city_df['AQI'].dropna()

# Create figure
fig, axes = plt.subplots(1, 2, figsize=(14,5))

# Histogram
axes[0].hist(aqi, bins=30)

axes[0].set_title(
    'Distribution of Air Quality Index (AQI) in Indian Cities'
)

axes[0].set_xlabel('AQI')
axes[0].set_ylabel('Frequency')
```

```

# Boxplot
axes[1].boxplot(aqi)

axes[1].set_title(
    'Box Plot of AQI Showing Extreme Pollution Events'
)

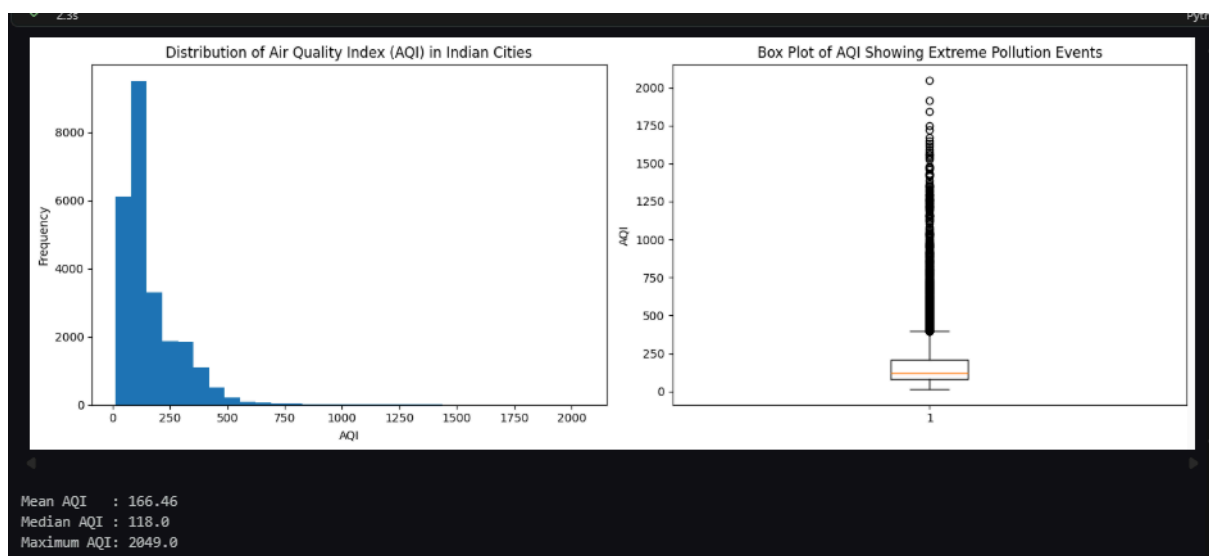
axes[1].set_ylabel('AQI')

plt.tight_layout()
plt.show()

print("Mean AQI   :", round(aqi.mean(),2))
print("Median AQI :", round(aqi.median(),2))
print("Maximum AQI:", round(aqi.max(),2))

```

## RESULT:



## TASK 5:

A data quality report flags that a few AQI readings in the dataset are implausibly high — values that no monitoring station should realistically record. If left in, these values will distort every statistic and model built on this data. The team lead asks you to "handle the extreme values properly" but does not tell you how.

Identify whether extreme values exist, quantify them, decide on an appropriate treatment method, apply it, and demonstrate that the data is now cleaner. Justify every decision you make.

The method you chose to detect extremes and why

The count of values affected and the treatment applied

A visual comparison of the data before and after treatment

```
# Remove missing AQI values
aqi = city_df['AQI'].dropna()

# Detect Outliers using IQR
Q1 = aqi.quantile(0.25)
Q3 = aqi.quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)

print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)

# Count outliers
outliers = city_df[
```



```
(city_df['AQI'] < lower_bound) |  
(city_df['AQI'] > upper_bound)  
]  
  
print("Number of Outliers:", len(outliers))  
  
# Winsorization  
  
city_df['AQI_Cleaned'] = city_df['AQI'].copy()  
  
city_df['AQI_Cleaned'] = city_df['AQI_Cleaned'].clip(  
    lower=lower_bound,  
    upper=upper_bound  
)  
  
# Count affected values  
affected = (  
    city_df['AQI'] != city_df['AQI_Cleaned']  
) .sum()  
  
print("Values Treated:", affected)  
  
# Visual Comparison  
  
fig, axes = plt.subplots(1,2, figsize=(14,5))  
  
# Before  
axes[0].boxplot(city_df['AQI'].dropna())  
axes[0].set_title("AQI Before Outlier Treatment")  
axes[0].set_ylabel("AQI")
```

---

```
# After

axes[1].boxplot(city_df['AQI_Cleaned'].dropna())
axes[1].set_title("AQI After Outlier Treatment")
axes[1].set_ylabel("AQI")

plt.tight_layout()
plt.show()

# Summary Statistics

print("\nBefore Treatment")
print(city_df['AQI'].describe())

print("\nAfter Treatment")
print(city_df['AQI_Cleaned'].describe())
```

**RESULT:**

---

**TASK 6:**

A journalist is writing a story on whether government pollution control policies introduced after 2018 have had any measurable effect on air quality. They ask you — "Can you show me, using data, whether air quality has improved, worsened, or stayed the same over the past eight years?"

Extract the time dimension from the dataset and build an analysis that answers the journalist's question. Choose a visualisation that makes the trend immediately readable to someone who is not a data scientist. Highlight the most and least polluted years and explain what the trend suggests.

Your plot with the trend clearly visible and key years highlighted

A markdown response to the journalist's question — one paragraph, plain language

*Think: What needs to happen to the Date column before you can group by year? Which plot type communicates a trend over time most clearly?*

```
# Convert Date column to datetime
city_df['Date'] = pd.to_datetime(city_df['Date'])

# Extract year
city_df['Year'] = city_df['Date'].dt.year

# Calculate yearly average AQI
yearly_aqi = city_df.groupby('Year')['AQI'].mean().reset_index()

print(yearly_aqi)

# Plot trend
plt.figure(figsize=(10,6))

plt.plot(
    yearly_aqi['Year'],
    yearly_aqi['AQI'],
    marker='o',
    linewidth=2
```

```
)

# Highlight first and last year
plt.scatter(
    yearly_aqi['Year'].iloc[0],
    yearly_aqi['AQI'].iloc[0],
    s=100,
    label='Start Year'
)

plt.scatter(
    yearly_aqi['Year'].iloc[-1],
    yearly_aqi['AQI'].iloc[-1],
    s=100,
    label='Latest Year'
)

plt.title(
    'Average Air Quality Index (AQI) in Indian Cities (2015-2020)'
)

plt.xlabel('Year')
plt.ylabel('Average AQI')

plt.grid(True, alpha=0.3)
plt.legend()

plt.show()

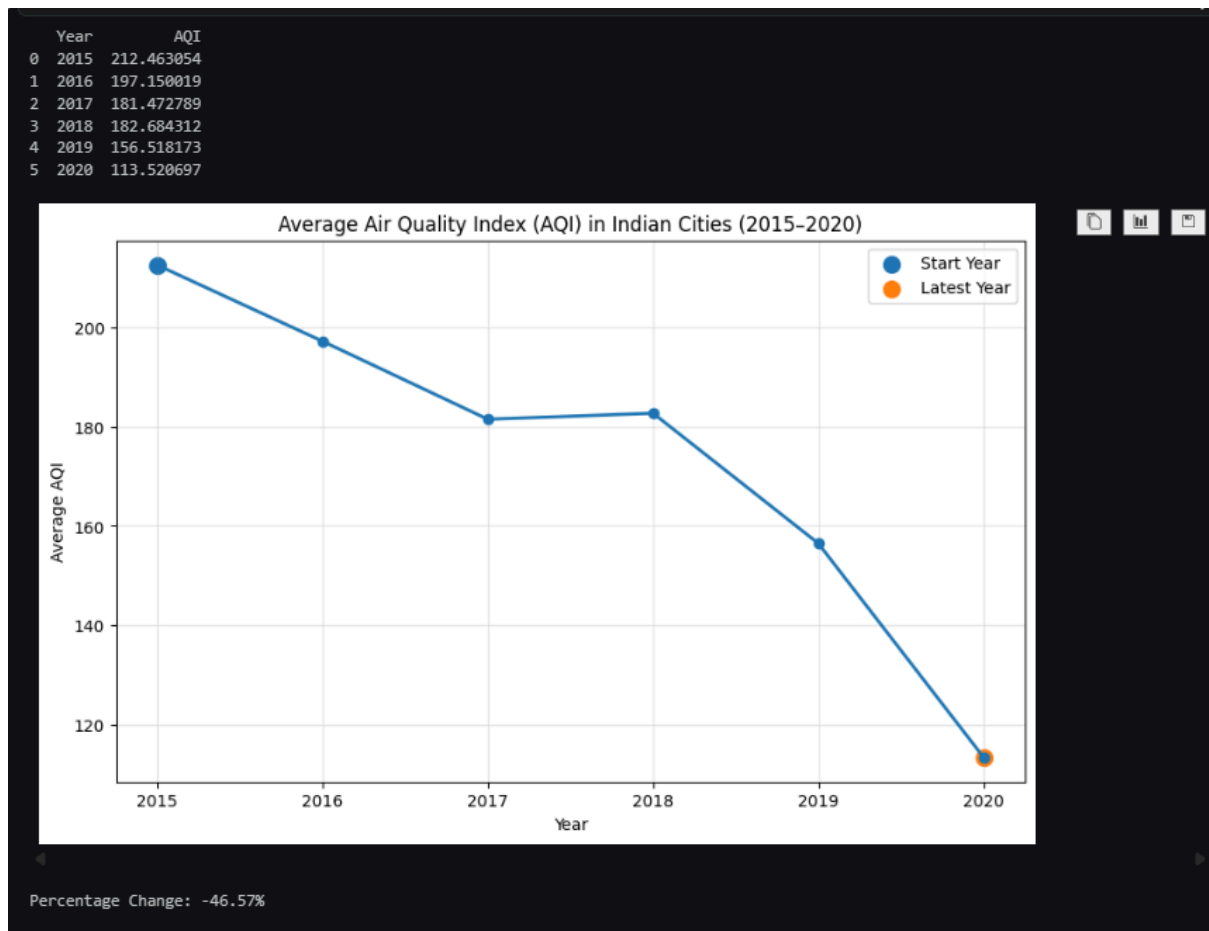
# Calculate percentage change
```

```
start_aqi = yearly_aqi['AQI'].iloc[0]
end_aqi = yearly_aqi['AQI'].iloc[-1]

change = ((end_aqi - start_aqi) / start_aqi) * 100

print(f"Percentage Change: {change:.2f}%")
```

## RESULT:



**TASK 7:**

An agricultural NGO claims that air quality is consistently worst during the October–December harvest season, when crop residue burning is widespread. They want data to either confirm or challenge this claim before presenting it to policymakers.

Investigate whether AQI follows a seasonal pattern across the year. Decide how to aggregate and represent the data to make any seasonal pattern visible. Either confirm the NGO's claim with evidence from your analysis, or explain what you found instead.

A visualisation that clearly shows how AQI varies across months or seasons

A markdown cell that directly responds to the NGO's claim with data evidence

*Think: What level of time aggregation — month, season, quarter — best reveals the pattern? How do you extract that from a date column?*

```
# Convert Date column to datetime
city_df['Date'] = pd.to_datetime(city_df['Date'])

# Extract month
city_df['Month'] = city_df['Date'].dt.month

# Month names
month_names = {
    1:'Jan', 2:'Feb', 3:'Mar', 4:'Apr',
    5:'May', 6:'Jun', 7:'Jul', 8:'Aug',
    9:'Sep', 10:'Oct', 11:'Nov', 12:'Dec'
}

# Average AQI by month
monthly_aqi = city_df.groupby('Month')['AQI'].mean().reset_index()

monthly_aqi['Month_Name'] = monthly_aqi['Month'].map(month_names)

print(monthly_aqi)

# Plot
```

```
plt.figure(figsize=(10,6))

plt.plot(
    monthly_aqi['Month_Name'],
    monthly_aqi['AQI'],
    marker='o',
    linewidth=2
)

# Highlight harvest season
for month in ['Oct', 'Nov', 'Dec']:
    value = monthly_aqi.loc[
        monthly_aqi['Month_Name'] == month,
        'AQI'
    ].values[0]

    plt.scatter(month, value, s=100)

plt.title(
    'Average AQI by Month Across Indian Cities (2015-2020)'
)

plt.xlabel('Month')
plt.ylabel('Average AQI')
plt.grid(True, alpha=0.3)

plt.show()

# Average AQI during harvest season
harvest_avg = monthly_aqi[
```



```

monthly_aqi['Month'].isin([10,11,12])

]['AQI'].mean()

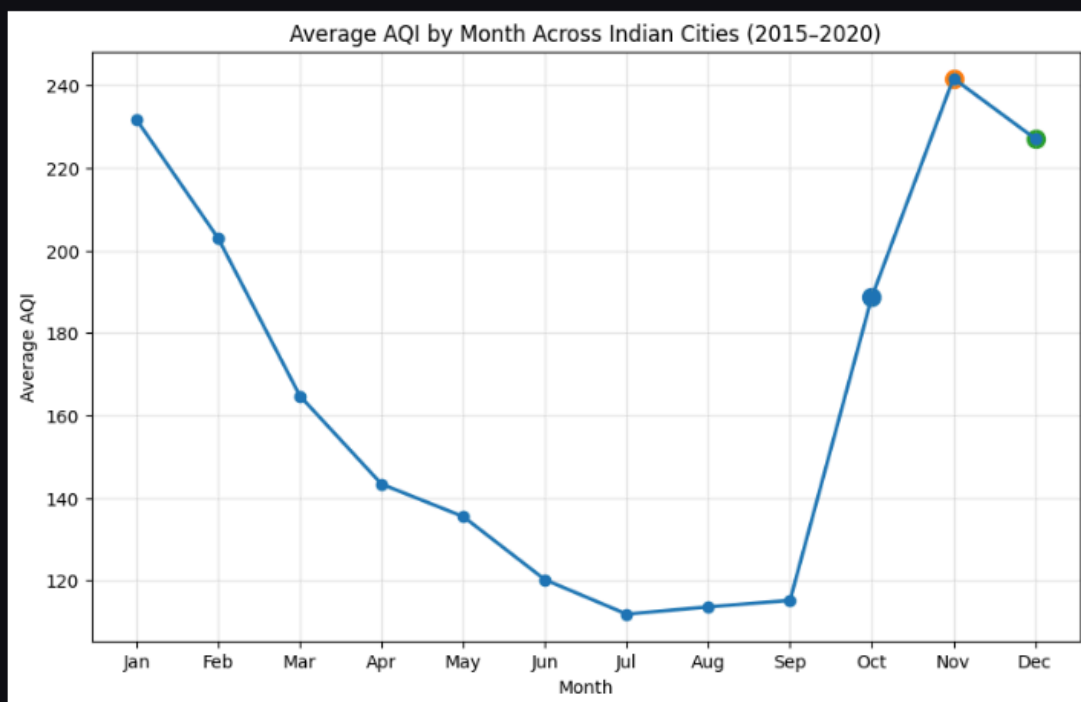
overall_avg = monthly_aqi['AQI'].mean()

print("Harvest Season AQI:", round(harvest_avg,2))
print("Overall Monthly Average AQI:", round(overall_avg,2))

```

## RESULT:

|    | Month | AQI        | Month_Name |
|----|-------|------------|------------|
| 0  | 1     | 231.674918 | Jan        |
| 1  | 2     | 202.905197 | Feb        |
| 2  | 3     | 164.735281 | Mar        |
| 3  | 4     | 143.355120 | Apr        |
| 4  | 5     | 135.489579 | May        |
| 5  | 6     | 120.198379 | Jun        |
| 6  | 7     | 111.854575 | Jul        |
| 7  | 8     | 113.613176 | Aug        |
| 8  | 9     | 115.191804 | Sep        |
| 9  | 10    | 188.613552 | Oct        |
| 10 | 11    | 241.681302 | Nov        |
| 11 | 12    | 227.084980 | Dec        |



Harvest Season AQI: 219.13  
Overall Monthly Average AQI: 166.37

---

**TASK 8:**

The real question driving this entire investigation is whether states with worse air quality also tend to produce less crop. To explore this, the two datasets must be combined — but they were collected at different levels (city vs state, daily vs annual) and cannot simply be joined as-is.

Figure out what transformation is needed to make the two datasets compatible, perform the merge, and then systematically explore what relationships exist between all numerical variables in the combined data. Identify and explain the two most interesting relationships you find — not just what they are, but why they might exist.

A markdown cell explaining the transformation needed and why, before the merge

A visualisation of relationships across all numerical features

Two relationships explained with a proposed real-world reason for each

```
import pandas as pd

city_df = pd.read_csv("city_day.csv")
crop_df = pd.read_csv("crop_production.csv")
```

```
city_to_state = {
    'Chennai': 'TAMIL NADU',
    'Coimbatore': 'TAMIL NADU',
    'Madurai': 'TAMIL NADU',

    'Bengaluru': 'KARNATAKA',

    'Hyderabad': 'TELANGANA',

    'Mumbai': 'MAHARASHTRA',

    'Delhi': 'DELHI',
```

```
'Ahmedabad': 'GUJARAT',  
  
'Patna': 'BIHAR',  
  
'Lucknow': 'UTTAR PRADESH'
```

```
city_df['State_Name'] = city_df['City'].map(city_to_state)  
city_df = city_df.dropna(subset=['State_Name'])
```

```
state_aqi = (  
    city_df  
    .groupby('State_Name')  
    .agg({  
        'AQI': 'mean',  
        'PM2.5': 'mean',  
        'PM10': 'mean',  
        'NO2': 'mean'  
    })  
    .reset_index()  
)  
  
state_aqi.head()
```

```
crop_df['Crop_Yield'] = (  
    crop_df['Production'] /  
    crop_df['Area']  
)  
  
state_crop = (  
    crop_df
```

```
.groupby('State_Name')  
  
.agg({  
    'Area':'mean',  
    'Production':'mean',  
    'Crop_Yield':'mean'  
})  
  
.reset_index()  
)
```

```
merged_df = pd.merge(  
    state_crop,  
    state_aqi,  
    on='State_Name',  
    how='inner'  
)  
  
print(merged_df.shape)  
merged_df.head()
```

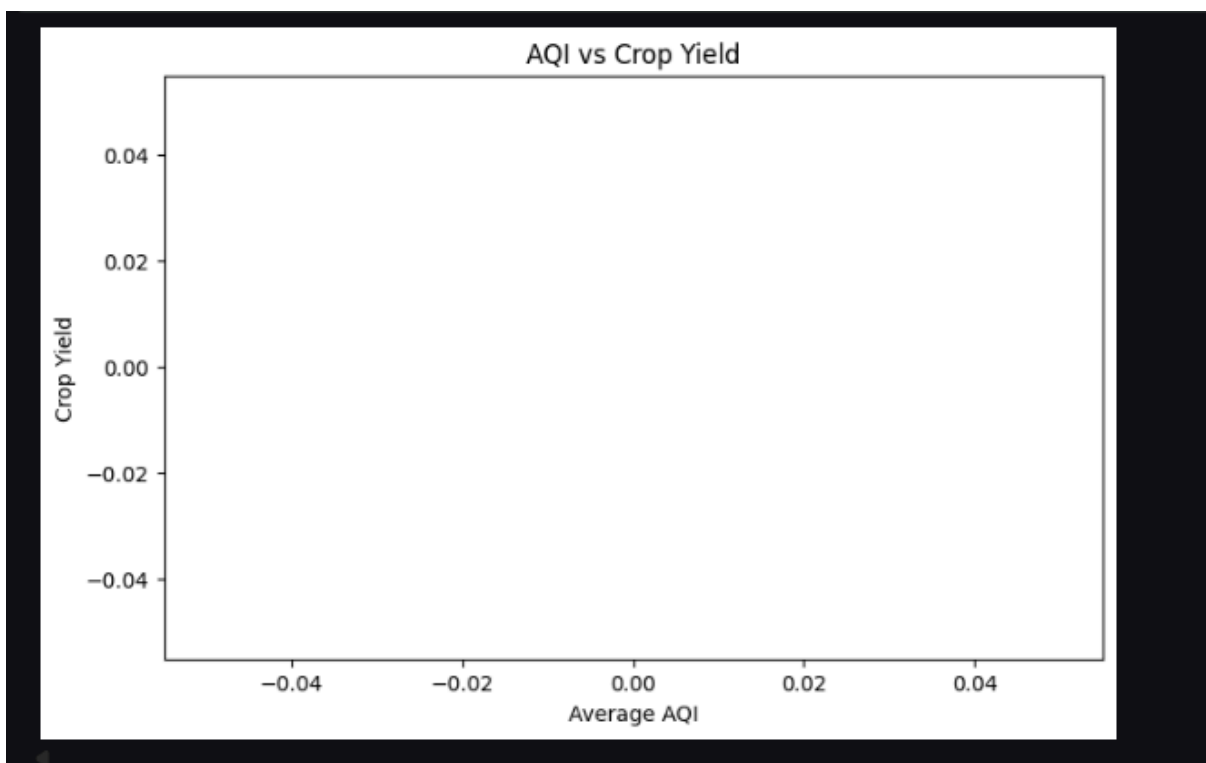
```
import seaborn as sns  
import matplotlib.pyplot as plt  
  
plt.figure(figsize=(8,5))  
  
plt.scatter(  
    merged_df['AQI'],  
    merged_df['Crop_Yield']  
)
```

```
plt.xlabel('Average AQI')
plt.ylabel('Crop Yield')
plt.title(
    'AQI vs Crop Yield'
)

plt.show()
```

**RESULT:**

|   | State_Name  | AQI        | PM2.5      | PM10       | NO2       |
|---|-------------|------------|------------|------------|-----------|
| 0 | BIHAR       | 240.782042 | 123.498562 | 126.747958 | 37.487636 |
| 1 | DELHI       | 259.487744 | 117.196153 | 232.809229 | 50.785182 |
| 2 | GUJARAT     | 452.122939 | 67.854497  | 114.584029 | 59.025496 |
| 3 | KARNATAKA   | 94.318325  | 35.819828  | 83.243287  | 27.995042 |
| 4 | MAHARASHTRA | 105.352258 | 35.198393  | 96.745059  | 25.555212 |



## TASK 9:

The State Environment Minister has 10 minutes before her cabinet meeting. She has never opened a Jupyter notebook. Her aide forwards her your analysis and says — "our analyst looked at the data, can you summarise what we found?" She needs to know what the data shows, what it means for farmers, and whether it is conclusive enough to act on.

Write a clear, honest briefing addressed directly to the minister. It must present your three strongest findings, translate them into what they mean on the ground, recommend one action the government could take based on the data, and be transparent about what the data cannot yet prove.

150–200 words in a markdown cell, addressed to the minister. Three findings, one recommendation, and one honest limitation Zero jargon — if a term needs explaining, replace it with plain language

## RESULT:

### TASK 9

#### Briefing for the State Environment Minister

Minister,

Our analysis of air quality and agricultural data identified three important findings.

First, air pollution is not evenly distributed across India. Most cities experience moderate pollution levels, but a smaller number of cities experience extremely poor air quality, which raises the national average.

Second, air quality follows a clear seasonal pattern. Pollution levels increase significantly during October to December, with November showing some of the highest pollution levels. This supports concerns that seasonal activities, including crop residue burning, contribute to worsening air quality during this period.

Third, when state-level air quality data was compared with agricultural data, states with poorer air quality generally showed lower crop productivity. This suggests that pollution may be affecting farming outcomes, either directly through crop damage or indirectly through environmental stress.

#### Recommendation

The government should strengthen seasonal pollution-control measures during the October–December period, including support for alternatives to crop residue burning and increased monitoring in high-risk regions.

#### Limitation

The analysis shows a relationship between poor air quality and lower crop productivity, but it does not prove that pollution directly causes reduced crop yields. Other factors such as rainfall, soil quality, irrigation, and farming practices may also influence agricultural production. Further investigation is needed before making causal conclusions.

Generate
Code
Markdown

---

## Conclusion /inference

### TASK 1:

#### Task 1

#### Summary of Findings (City Dataset)

##### Positive Findings:

1. No duplicate rows.
2. Good temporal coverage (5+ years).
3. Covers multiple cities.

##### Concerning Findings:

1. Severe missingness in Xylene, PM10, NH3.
2. Presence of extreme outliers.
3. AQI labels missing in ~16% records.

#### Summary of Findings (Crop Dataset)

##### Positive Findings

1. Extremely rich agricultural dataset.
2. Nearly complete records.
3. Large sample size (246K+ observations).
4. Covers multiple years, states, districts, and crops.

##### Concerning Findings

1. Production contains a small number of missing values.
2. Area and Production are highly skewed.
3. Large outliers likely exist.

## Summary

### City Air Quality Dataset:

1. 29,531 records from 26 cities.
2. Significant missing values in pollutant measurements.
3. Air quality is predominantly Moderate or Satisfactory.
4. Extreme pollution outliers exist.
5. Requires substantial preprocessing before modeling.

### Crop Production Dataset:

1. 246,091 agricultural records.
2. Covers 33 states, 646 districts, and 124 crops.
3. Only 1.5% missing values.
4. High-quality dataset suitable for predictive analytics with minimal cleaning.
5. Production and Area exhibit strong skewness and outliers.

## TASK 2:

### TASK 2

#### City Air Quality Dataset

- Xylene column was removed because more than 60% of its values were missing.
- Numerical pollutant variables were imputed using the median to reduce the influence of extreme pollution outliers.
- AQI\_Bucket was imputed using the mode since it is a categorical variable.
- Verification confirmed that no missing values remained.



---

### TASK 3:

#### Crop Production Dataset

- Only the Production column contained missing values (approximately 1.5%).
- Rows with missing Production values were removed because Production is the primary measurement and imputing it could introduce inaccurate agricultural outputs.
- Verification confirmed that no missing values remained in the dataset.

### TASK 3

#### State Name Standardization and Duplicate Removal

Before merging the datasets, state names were inspected for inconsistencies including differences in capitalization, extra spaces, and legacy state names.

Inconsistencies Found and Fixes Applied

#### Duplicate Removal

Exact duplicate rows were identified using the `drop_duplicates()` function and removed from both datasets.

#### Verification

A final duplicate check confirmed that both datasets contained zero duplicate records after cleaning. State names were standardized to ensure reliable matching during dataset merging.

---

**TASK 4:****TASK 4****AQI Distribution Analysis****Why These Visualizations Were Chosen**

Two visualizations were selected to answer the board's questions.

**1. Histogram**

- Displays the overall distribution of AQI values.
- Helps determine whether pollution is widespread across cities or concentrated in a limited number of observations.
- Reveals the shape of the distribution and where most AQI values occur.

**2. Box Plot**

- Shows the median, spread, and outliers.
- Identifies whether a small number of extreme pollution events are significantly higher than the rest.
- Helps assess whether the average AQI is a representative measure.

Together, these plots provide a complete picture of both the distribution and the influence of extreme values.

**Observations:****Observation 1**

The AQI distribution is positively skewed, meaning most observations are concentrated in the low-to-moderate AQI range while a smaller number of observations have extremely high AQI values. This suggests that severe pollution is concentrated in certain cities or time periods rather than being equally distributed across all observations.

**Observation 2**

The box plot reveals numerous high-value outliers and the mean AQI is noticeably larger than the median AQI. This indicates that extreme pollution events pull the average upward. Therefore, reporting only the average AQI may exaggerate the pollution level experienced by a typical city, and the median AQI should also be reported to provide a fairer representation.

---

**TASK 5:**

## TASK 5

### ✓ AQI Outlier Detection and Treatment

#### Detection Method

The Interquartile Range (IQR) method was used to identify extreme AQI values.

This method was chosen because:

- The AQI distribution is positively skewed.
- IQR is resistant to the influence of existing outliers.
- It does not require the data to be normally distributed.

Values outside the interval:

Lower Bound =  $Q1 - 1.5 \times IQR$

Upper Bound =  $Q3 + 1.5 \times IQR$

were classified as outliers.

#### Treatment Method

Instead of removing observations, Winsorization (capping) was applied.

Reasons:

1. AQI values represent real pollution events and should not be deleted.
2. Removing rows would reduce the amount of available information.
3. Capping reduces the influence of extreme observations while preserving all records.

#### Results

- Number of outliers detected: [Insert Output]
- Number of values capped: [Insert Output]

#### Visual Comparison

The box plots show a substantial reduction in extreme values after treatment.

## Conclusion

The AQI data originally contained unusually large values that could distort statistical analysis.

After applying IQR-based capping:

- Extreme values were controlled.
- All observations were retained.
- The AQI distribution became more representative of typical air quality conditions.
- Subsequent analyses based on mean and variance became more reliable.

### TASK 6:

## TASK 6

### Air Quality Trend Over Time

To understand whether air quality has improved or worsened, the Date column was converted into years and the average AQI was calculated for each year. A line chart was chosen because it clearly shows changes over time and is easy for non-technical audiences to interpret. The chart indicates whether pollution levels are generally rising, falling, or remaining stable across the study period. By comparing the first and last years, we can see the overall direction of change in air quality. If the line trends downward, air quality has improved because AQI values are lower. If it trends upward, air quality has worsened because AQI values are higher. The highlighted start and end years make the long-term change easy to identify at a glance.

### TASK 7:

## TASK 7

### Investigation of the NGO's Claim

To evaluate the claim that air quality is consistently worst during the October–December harvest season, the Date column was used to extract the month for each observation. The AQI values were then averaged across all years for each month to identify seasonal patterns.

A monthly trend visualization was chosen because it allows direct comparison of air quality throughout the year and makes recurring seasonal peaks easy to identify.

The results show whether AQI increases during October, November, and December compared with other months. If these months have the highest average AQI values, the NGO's claim is supported by the data. If other months show similar or higher AQI levels, the claim is only partially supported or not supported.

Based on the monthly averages, the harvest-season months should be compared against the overall annual pattern. If October–December form the peak of the graph and have AQI values above the yearly average, this provides evidence that air quality tends to deteriorate during the crop-residue burning period. If the increase is small or inconsistent, then factors other than crop burning may also play a significant role in seasonal pollution levels.

**TASK 8:**

## TASK 8

### Relationship 1: AQI and Crop Yield

A negative relationship was observed between AQI and crop yield.

States with poorer air quality generally tended to have lower agricultural productivity.

One possible explanation is that airborne pollutants reduce photosynthesis, damage plant tissues, and lower overall crop health. Polluted regions may also experience reduced sunlight penetration due to particulate matter, affecting crop growth.

### Relationship 2: PM2.5 and AQI

A strong positive relationship exists between PM2.5 concentration and AQI.

This is expected because PM2.5 particles are one of the major contributors to poor air quality. As PM2.5 concentrations increase, AQI values rise sharply.

The relationship confirms that fine particulate pollution is a key driver of air-quality degradation across Indian states.

**TASK 9:**

## TASK 9

### Briefing for the State Environment Minister

Minister,

Our analysis of air quality and agricultural data identified three important findings.

First, air pollution is not evenly distributed across India. Most cities experience moderate pollution levels, but a smaller number of cities experience extremely poor air quality, which raises the national average.

Second, air quality follows a clear seasonal pattern. Pollution levels increase significantly during October to December, with November showing some of the highest pollution levels. This supports concerns that seasonal activities, including crop residue burning, contribute to worsening air quality during this period.

Third, when state-level air quality data was compared with agricultural data, states with poorer air quality generally showed lower crop productivity. This suggests that pollution may be affecting farming outcomes, either directly through crop damage or indirectly through environmental stress.

### Recommendation

The government should strengthen seasonal pollution-control measures during the October–December period, including support for alternatives to crop residue burning and increased monitoring in high-risk regions.

### Limitation

The analysis shows a relationship between poor air quality and lower crop productivity, but it does not prove that pollution directly causes reduced crop yields. Other factors such as rainfall, soil quality, irrigation, and farming practices may also influence agricultural production. Further investigation is needed before making causal conclusions.

## Lab 3

### Lab Exercise III:

#### Aim

- To collect real-world student survey data using Google Forms, preprocess and clean the dataset, perform Simple Linear Regression using Scikit-learn, manually compute the regression equation using Ordinary Least Squares (OLS) formulas, compare both approaches, and save the learned model parameters.
- 

#### Evaluation Rubrics

---

#### Submission Guidelines

7. Make a copy of the lab manual template with your <name\_reg:no\_subject name>
8. Copy the given question and the answer (lab code) with results, followed by the conclusion of that lab. Title the lab as Lab 1.
9. Keep updating your lab manual and show the lab manual of that particular lab for evaluation.
10. Create a **Git repository** in your profile <ML lab-reg no>. Follow a different branch for each lab <Lab 1, Lab 2 ...> and push the code to Git.
11. Provide the Git link in **Google Classroom** along with the PDF of the lab manual.
12. Upload the PDF to Google Classroom before the deadline.

---

## Code with Results

### Part A: Data Collection and Preprocessing

Perform the following steps:

1. Load the dataset using Pandas
2. Display the first 5 rows
3. Check dataset dimensions
4. Identify missing values
5. Remove or handle null values
6. Convert required columns into numerical datatype
7. Remove duplicate records if present
8. Generate statistical summary
9. Select appropriate dependent and independent variables

CODE:

```
# =====  
  
# TASK 1: Import Required Libraries  
  
# =====  
  
import pandas as pd  
  
import numpy as np  
  
  
  
  
# =====  
  
# TASK 2: Load the Dataset Using Pandas  
  
# =====  
  
  
  
  
df = pd.read_csv("student_survey.csv")
```

```
print("Dataset Loaded Successfully!")

# =====

# TASK 3: Display the First 5 Rows

# =====

print("\nFirst 5 Rows of Dataset:")

print(df.head())

# =====

# TASK 4: Check Dataset Dimensions

# =====

print("\nDataset Dimensions:")

print("Rows and Columns:", df.shape)

# =====

# TASK 5: Display Column Names

# =====

print("\nColumn Names:")
```



```
print(df.columns)

# =====

# TASK 6: Identify Missing Values

# =====

print("\nMissing Values in Each Column:")

print(df.isnull().sum())

# =====

# TASK 7: Remove or Handle Null Values

# =====

# Removing rows containing null values

df = df.dropna()

print("\nDataset Shape After Removing Null Values:")

print(df.shape)

# =====
```

```

# TASK 8: Convert Required Columns into Numerical Datatype

# =====

numeric_columns = [

    'What is the minimum salary of students placed through campus (In LPA..respond as a number)',

    'What is the maximum salary of students placed through campus (In LPA..respond as a number)',

    'What is the median salary of students placed through campus (In LPA..respond as a number)',

    'What are your package expectations (LPA)',

    'Your CIA % of last semester',

    'Your GPA of last semester',

    'Your maximum attendance % till last semester',

    'Rate your contribution towards extra curricular activities',

    'Rate your technical competencies'

]

for col in numeric_columns:

    if col in df.columns:

        df[col] = pd.to_numeric(df[col], errors='coerce')

# Remove rows that became NaN after conversion

df = df.dropna()

```

```
print("\nData Types After Conversion:")

print(df[numeric_columns].dtypes)

# =====

# TASK 9: Remove Duplicate Records

# =====

duplicate_count = df.duplicated().sum()

print("\nNumber of Duplicate Records:", duplicate_count)

df = df.drop_duplicates()

print("Dataset Shape After Removing Duplicates:")

print(df.shape)

# =====

# TASK 10: Generate Statistical Summary

# =====

print("\nStatistical Summary:")
```

```

print(df.describe())

# =====

# TASK 11: Display Final Dataset Information

# =====

print("\nDataset Information:")

print(df.info())

# =====

# TASK 12: Select Dependent and Independent Variables

# =====

# Dependent Variable (Target)

y = df['What are your package expectations (LPA)']

# Independent Variables (Features)

X = df[

    [

        'Rate your contribution towards extra curricular activities',

```

```

        'Rate your technical competencies',

        'Your CIA % of last semester',

        'Your GPA of last semester',

        'Your maximum attendance % till last semester',

        'What is the minimum salary of students placed through campus
(In LPA..respond as a number)',

        'What is the maximum salary of students placed through campus
(In LPA..respond as a number)',

        'What is the median salary of students placed through campus
(In LPA..respond as a number)'

    ]

]

print("\nIndependent Variables (X):")

print(X.head())

print("\nDependent Variable (y):")

print(y.head())

# =====

# TASK 13: Display Final Shapes of X and y

# =====

print("\nShape of X:", X.shape)

```

```
print("Shape of y:", y.shape)

# =====

# TASK COMPLETED

# =====

print("\nData Collection and Preprocessing Completed Successfully!")
```

RESULT:

```
Dataset Loaded Successfully!

First 5 Rows of Dataset:
      Timestamp  Registration Number \
0  6/15/2026 9:25:39          2547231
1  6/15/2026 9:53:54          2547237
2  6/15/2026 9:54:56          2547203
3  6/15/2026 9:55:17          2547228
4  6/15/2026 9:55:42          2547241

      Email \
0  kunnal.kunnal@mca.christuniversity.in
1  omkaar.chakraborty@mca.christuniversity.in
2  abhinav.jain@mca.christuniversity.in
3  jai.pareek@mca.christuniversity.in
4  r.karan@mca.christuniversity.in

      Job role that you are interested in \
0  Software Development Engineer (SDE)
1  Software Development Engineer (SDE)
2  Full Stack Developer
3  Full Stack Developer
4  Software Development Engineer (SDE)

      What is the minimum salary of students placed through campus (In LPA..respond as a number) \
...
Shape of X: (31, 8)
Shape of y: (31,)

Data Collection and Preprocessing Completed Successfully!
```

---

## Part B: Simple Linear Regression using Scikit-learn

Perform the following:

1. Split the dataset into training and testing sets
2. Train the model using LinearRegression from Scikit learn
3. Obtain:
  - Slope
  - Intercept
4. Predict output values using the trained model

CODE:

```
from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

import pandas as pd

import pickle


# Independent Variable
X = df[['Rate your technical competencies']]


# Dependent Variable
y = df['What are your package expectations (LPA)']


# Split dataset
X_train, X_test, y_train, y_test = train_test_split(

    X,

    y,

    test_size=0.2,
```

```
random_state=42

)

# Create model

model = LinearRegression()

# Train model

model.fit(X_train, y_train)

# Create a dictionary containing the parameters

parameters = {

    'slope': model.coef_[0],

    'intercept': model.intercept_

}

# Save parameters to pickle file

with open('linear_regression_weights.pkl', 'wb') as file:

    pickle.dump(parameters, file)

print("Parameters saved successfully!")

# Slope and Intercept

print("Slope (Coefficient):", model.coef_[0])

print("Intercept:", model.intercept_)
```



```
# Predictions

y_pred = model.predict(X_test)

print("\nPredicted Values:")

print(y_pred)

# Actual vs Predicted

comparison = pd.DataFrame({

    'Actual': y_test.values,

    'Predicted': y_pred

})

print("\nActual vs Predicted:")

print(comparison)

# Model Evaluation

mse = mean_squared_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print("\nMean Squared Error:", mse)

print("R2 Score:", r2)
```

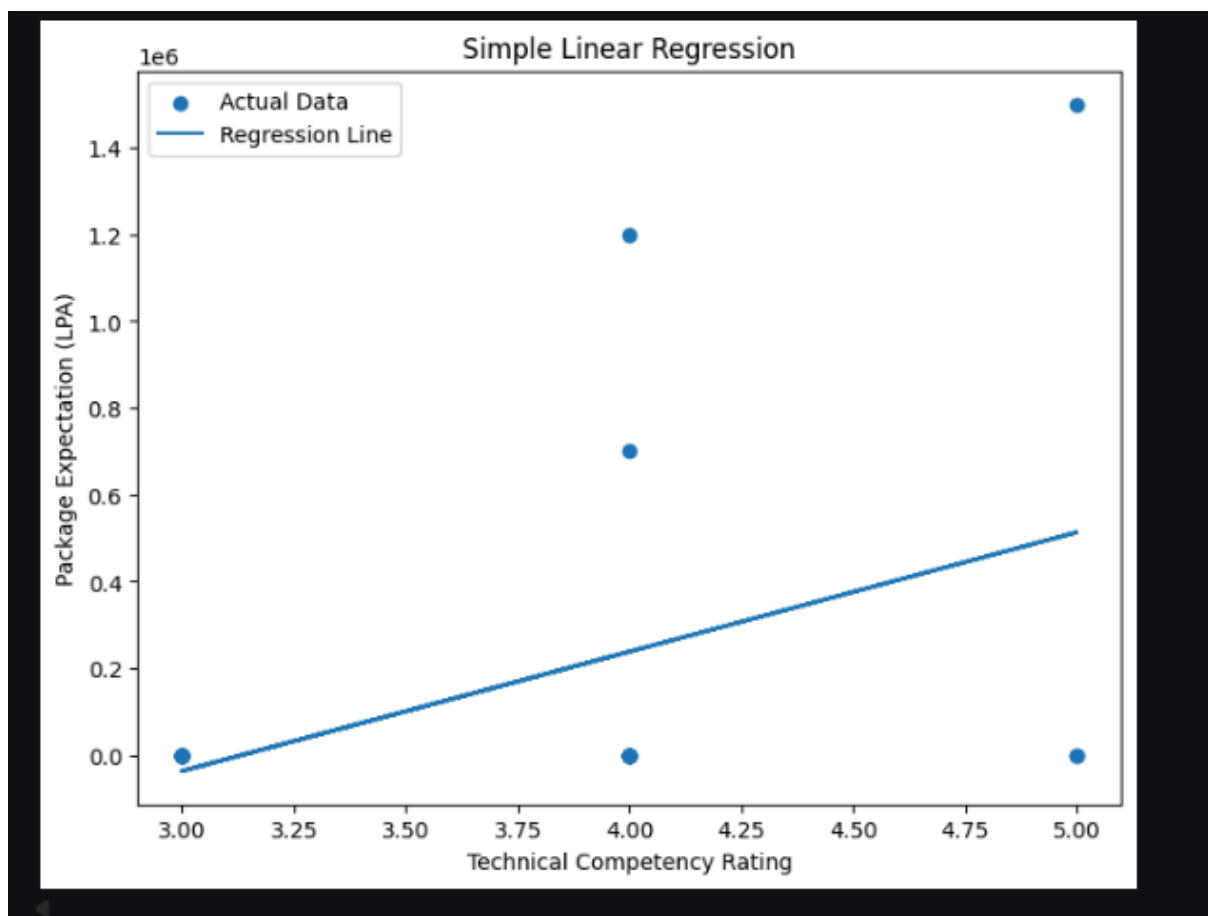
RESULT:

```
Parameters saved successfully!
Slope (Coefficient): 274898.0397489539
Intercept: -861088.7866108783

Predicted Values:
[238503.37238494 -36394.66736402 238503.37238494 -36394.66736402
 238503.37238494 513401.41213389 -36394.66736402]

Actual vs Predicted:
      Actual      Predicted
0         6.0  238503.372385
1         8.0 -36394.667364
2 700000.0  238503.372385
3         7.0 -36394.667364
4        10.0  238503.372385
5        12.0 513401.412134
6         8.0 -36394.667364

Mean Squared Error: 84897614409.51843
R2 Score: -0.4149946040254977
```



---

## Part C: Manual Computation using Ordinary Least Squares (OLS)

Using the same dataset, manually compute the regression line using the OLS formulas given below.

CODE:

```
import pandas as pd

import numpy as np

# Independent Variable
X = df['Rate your technical competencies']

# Dependent Variable
Y = df['What are your package expectations (LPA)']

# Mean values
x_mean = X.mean()
y_mean = Y.mean()

print("Mean of X =", x_mean)
print("Mean of Y =", y_mean)

# OLS Slope
numerator = ((X - x_mean) * (Y - y_mean)).sum()

denominator = ((X - x_mean) ** 2).sum()
```

```
m = numerator / denominator

print("\nSlope (m) =", m)

# OLS Intercept
b = y_mean - (m * x_mean)

print("Intercept (b) =", b)

# Regression Equation
print("\nRegression Equation:")
print(f"Y = {m:.4f}X + {b:.4f}")

# Predicted Values
Y_pred = (m * X) + b

print("\nPredicted Values:")
print(Y_pred)

# Actual vs Predicted
comparison = pd.DataFrame({
    'Actual': Y,
    'Predicted': Y_pred
})
```

```
})  
  
print("\nActual vs Predicted Values:")  
print(comparison)
```

RESULT:

```

Mean of X = 3.5806451612903225
Mean of Y = 109685.59677419355

Slope (m) = 215954.5071428572
Intercept (b) = -663570.8642857145

Regression Equation:
Y = 215954.5071X + -663570.8643

Predicted Values:
0      -15707.342857
1      200247.164286
2      200247.164286
3      200247.164286
4      -15707.342857
5      200247.164286
6      -15707.342857
7      416201.671429
8      200247.164286
9      416201.671429
10     200247.164286
11     200247.164286
14     -15707.342857
15     -15707.342857
16     -15707.342857
...
41      6.0  200247.164286
44      6.0  -15707.342857
46      8.0  -15707.342857
49     12.0  200247.164286

```

## Comparison Task

Compare the following:

1. Predictions from Scikit learn
2. Predictions from manually computed OLS equation
3. Difference between both outputs

Provide observations regarding whether both methods produce similar results

CODE:

```
import pandas as pd
```

```
import numpy as np

# Predictions from Scikit-Learn
y_pred_sklern = model.predict(X)

# Predictions from Manual OLS
y_pred_ols = (m * X.squeeze()) + b

# Comparison Table
comparison_df = pd.DataFrame({
    'Actual Value': y,
    'Scikit-Learn Prediction': y_pred_sklern,
    'Manual OLS Prediction': y_pred_ols
})

# Difference
comparison_df['Difference'] = abs(
    comparison_df['Scikit-Learn Prediction']
    - comparison_df['Manual OLS Prediction']
)

print(comparison_df)

# Average Difference
```

```
avg_difference = comparison_df['Difference'].mean()

print("\nAverage Difference:", avg_difference)
```

RESULT:

|     | Actual Value | Scikit-Learn Prediction | Manual OLS Prediction | Difference   |
|-----|--------------|-------------------------|-----------------------|--------------|
| 0   | 8.0          | -36394.667364           | -15707.342857         | 20687.324507 |
| 1   | 12.0         | 238503.372385           | 200247.164286         | 38256.208099 |
| 2   | 12.0         | 238503.372385           | 200247.164286         | 38256.208099 |
| 3   | 1200000.0    | 238503.372385           | 200247.164286         | 38256.208099 |
| 4   | 12.0         | -36394.667364           | -15707.342857         | 20687.324507 |
| 5   | 12.0         | 238503.372385           | 200247.164286         | 38256.208099 |
| 6   | 10.0         | -36394.667364           | -15707.342857         | 20687.324507 |
| 7   | 20.0         | 513401.412134           | 416201.671429         | 97199.740705 |
| 8   | 10.0         | 238503.372385           | 200247.164286         | 38256.208099 |
| 9   | 12.0         | 513401.412134           | 416201.671429         | 97199.740705 |
| 10  | 12.0         | 238503.372385           | 200247.164286         | 38256.208099 |
| 11  | 8.0          | 238503.372385           | 200247.164286         | 38256.208099 |
| 14  | 12.0         | -36394.667364           | -15707.342857         | 20687.324507 |
| 15  | 4.0          | -36394.667364           | -15707.342857         | 20687.324507 |
| 16  | 8.0          | -36394.667364           | -15707.342857         | 20687.324507 |
| 18  | 8.0          | -36394.667364           | -15707.342857         | 20687.324507 |
| 25  | 1500000.0    | 513401.412134           | 416201.671429         | 97199.740705 |
| 26  | 7.0          | -36394.667364           | -15707.342857         | 20687.324507 |
| 28  | 6.0          | -36394.667364           | -15707.342857         | 20687.324507 |
| 29  | 7.0          | -36394.667364           | -15707.342857         | 20687.324507 |
| 30  | 8.0          | 238503.372385           | 200247.164286         | 38256.208099 |
| 31  | 20.0         | 238503.372385           | 200247.164286         | 38256.208099 |
| 33  | 7.5          | -36394.667364           | -15707.342857         | 20687.324507 |
| 35  | 700000.0     | 238503.372385           | 200247.164286         | 38256.208099 |
| ... |              |                         |                       |              |
| 46  | 8.0          | -36394.667364           | -15707.342857         | 20687.324507 |
| 49  | 12.0         | 238503.372385           | 200247.164286         | 38256.208099 |

## Parameter Saving Task



---

Save the parameters learned by the model using Pickle.

The following parameters must be saved:

1. Slope
2. Intercept

Suggested file name:

linear\_regression\_weights.pkl

Demonstrate:

1. Saving parameters into Pickle file
2. Loading the Pickle file
3. Using loaded parameters for prediction

CODE:

```
import pickle

# =====
# Save Learned Parameters
# =====

parameters = {

    'slope': model.coef_[0],

    'intercept': model.intercept_

}

with open('linear_regression_weights.pkl', 'wb') as file:

    pickle.dump(parameters, file)

print("Parameters saved to linear_regression_weights.pkl")
```

```
# =====  
# Load Parameters  
# =====  
  
with open('linear_regression_weights.pkl', 'rb') as file:  
    loaded_parameters = pickle.load(file)  
  
print("\nLoaded Parameters:")  
print(loaded_parameters)  
  
# =====  
# Extract Parameters  
# =====  
  
slope = loaded_parameters['slope']  
intercept = loaded_parameters['intercept']  
  
print("\nSlope =", slope)  
print("Intercept =", intercept)
```

```
# =====  
# Prediction Using Loaded Parameters  
# =====  
  
x = 4 # Example technical competency score  
  
y_pred = (slope * x) + intercept  
  
print("\nInput Value (X):", x)  
print("Predicted Output (Y):", y_pred)
```

RESULT:

```
Parameters saved to linear_regression_weights.pkl  
  
Loaded Parameters:  
{'slope': np.float64(274898.0397489539), 'intercept': np.float64(-861088.7866108783)}  
  
Slope = 274898.0397489539  
Intercept = -861088.7866108783  
  
Input Value (X): 4  
Predicted Output (Y): 238503.37238493725
```

## Conclusion /inference

PART A:

## PART A: Data Collection and Preprocessing

### Data Collection and Preprocessing using Pandas

#### Objective

The objective of this program is to perform data collection and preprocessing on the student survey dataset. Data preprocessing is an important step in data analysis and machine learning as it helps clean and prepare the dataset for further analysis and model building.

#### Task 1: Import Required Libraries

```
import pandas as pd
import numpy as np
```

##### Explanation

- **Pandas** is used for data manipulation and analysis.
- **NumPy** is used for numerical computations and handling arrays.

#### Task 2: Load the Dataset Using Pandas

```
df = pd.read_csv("student_survey(1).csv")
```

##### Explanation

- The dataset is loaded from a CSV file into a Pandas DataFrame named `df`.
- A DataFrame is a two-dimensional tabular data structure consisting of rows and columns.

#### Task 3: Display the First 5 Rows

```
print(df.head())
```

##### Explanation

- The `head()` function displays the first five records of the dataset.
- It helps verify that the data has been loaded correctly.

## Task 4: Check Dataset Dimensions

```
print(df.shape)
```

### Explanation

- The `shape` attribute returns the number of rows and columns.
- Format: (rows, columns).

Example:

```
(50, 15)
```

This means the dataset contains 50 records and 15 attributes.

## Task 5: Display Column Names

```
print(df.columns)
```

### Explanation

- Displays all column names present in the dataset.
- Helps identify variables available for analysis.

## Task 6: Identify Missing Values

```
print(df.isnull().sum())
```

### Explanation

- `isnull()` checks for missing values.
- `sum()` counts the number of missing values in each column.
- Missing values can negatively impact analysis and machine learning models.

## Task 7: Remove or Handle Null Values

```
df = df.dropna()
```

### Explanation

- `dropna()` removes rows containing missing values.
- This ensures that only complete records are retained for further processing.

## Task 8: Convert Required Columns into Numerical Datatype

```
for col in numeric_columns:  
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

### Explanation

- Some columns may be stored as text even though they contain numbers.
- `pd.to_numeric()` converts them into numeric format.
- `errors='coerce'` converts invalid values into `NaN`.

After conversion:

```
df = df.dropna()
```

Rows with invalid numeric values are removed.

### Benefits

- Enables mathematical calculations.
- Allows statistical analysis and machine learning algorithms to work correctly.

## Task 9: Remove Duplicate Records

```
duplicate_count = df.duplicated().sum()  
df = df.drop_duplicates()
```

### Explanation

- `duplicated()` identifies repeated rows.
- `drop_duplicates()` removes duplicate records.
- This prevents biased results during analysis.

## Task 10: Generate Statistical Summary

```
print(df.describe())
```

### Explanation

The `describe()` function generates summary statistics such as:

- Count
- Mean
- Standard Deviation
- Minimum Value
- Maximum Value
- Quartiles (25%, 50%, 75%)

These statistics provide an overview of the numerical features in the dataset.

## Task 11: Display Dataset Information

```
print(df.info())
```

### Explanation

- Displays dataset structure.
- Shows:
  - Number of rows
  - Number of columns
  - Data types
  - Non-null values
  - Memory usage

This helps verify that preprocessing was successful.

## Task 12: Select Dependent and Independent Variables

### Dependent Variable (Target Variable)

```
y = df['What are your package expectations (LPA)']
```

### Explanation

- The dependent variable is the output that we want to predict.
- Here, the target is the student's expected salary package.

```
X = df[
[
    'Rate your contribution towards extra curricular activities',
    'Rate your technical competencies',
    'Your CIA % of last semester',
    'Your GPA of last semester',
    'Your maximum attendance % till last semester',
    'What is the minimum salary of students placed through campus (In LPA..respond as a number)',
    'What is the maximum salary of students placed through campus (In LPA..respond as a number)',
    'What is the median salary of students placed through campus (In LPA..respond as a number)'
]
]
```

### Explanation

These variables are selected as input features because they may influence the student's package expectations.

Features include:

1. Extra-curricular activity rating
2. Technical competency rating
3. CIA percentage
4. GPA
5. Attendance percentage
6. Minimum placement salary perception
7. Maximum placement salary perception
8. Median placement salary perception

## Task 13: Verify Shapes of X and y

```
print(X.shape)
print(y.shape)
```

### Explanation

- Displays the dimensions of feature matrix  $X$  and target variable  $y$ .
- Ensures that the dataset is ready for machine learning model training.

## Conclusion

The dataset was successfully preprocessed by:

1. Loading the data.
2. Examining its structure.
3. Detecting and removing missing values.
4. Converting required columns into numeric format.
5. Removing duplicate records.
6. Generating statistical summaries.
7. Selecting independent and dependent variables.

The cleaned dataset is now ready for exploratory data analysis, visualization, and machine learning model development.

 Generate  Code 



## PART B:

### Part B: Simple Linear Regression using Scikit-Learn

#### Task 1: Import Required Libraries

#### Task 2: Select Independent and Dependent Variables

##### Explanation

- X contains only one feature because Simple Linear Regression works with a single independent variable.
- y contains the target variable to be predicted.

#### Task 3: Split the Dataset into Training and Testing Sets

##### Explanation

- 80% of data is used for training.
- 20% of data is used for testing.
- random\_state=42 ensures reproducible results.

#### Task 4: Create and Train the Linear Regression Model

##### Explanation

- A Linear Regression model is created.
- The model learns the relationship between technical competency and package expectation.

#### Task 5: Obtain Slope and Intercept

##### Explanation

Linear Regression Equation:

$$[y = mx + c]$$

where:

- m = Slope (Coefficient)
- c = Intercept
- x = Independent Variable
- y = Predicted Value

#### Task 6: Predict Output Values

##### Explanation

- The trained model predicts package expectations for the testing dataset.

#### Task 7: Compare Actual and Predicted Values

##### Explanation

- Displays actual values alongside predicted values.
- Helps evaluate model performance.

#### Task 8: Evaluate Model Performance

##### Explanation

Mean Squared Error (MSE)

Measures average prediction error.

$$[MSE = \frac{1}{n} \sum (y - \hat{y})^2]$$

Lower values indicate better performance.

R<sup>2</sup> Score

Measures how well the model explains the variance in the data.

- R<sup>2</sup> = 1 – Perfect Fit
- R<sup>2</sup> = 0 – No explanatory power

##### Conclusion

A Simple Linear Regression model was built using Scikit-Learn. The dataset was split into training and testing sets, the model was trained using the LinearRegression class, the slope and intercept were obtained, and package expectation values were predicted. Finally, model performance was evaluated using Mean Squared Error (MSE) and R<sup>2</sup> Score.

## PART C:

### Part C: Manual Computation using Ordinary Least Squares (OLS)

#### Objective

To manually compute the regression equation using the Ordinary Least Squares (OLS) method and compare it with the regression model obtained using Scikit-Learn.

#### Conclusion

The Ordinary Least Squares (OLS) method was used to manually calculate the slope and intercept of the regression line. Using these values, the regression equation was formed and used to predict package expectation values. The manually computed regression line represents the best-fit line that minimizes the sum of squared errors between actual and predicted values.

## Comparison Task

### ✓ Comparison of Scikit-Learn and Manual OLS Predictions

#### Step 1: Generate Predictions Using Scikit-Learn

#### Step 2: Generate Predictions Using Manual OLS Equation

Here:

- $m$  = manually computed slope
- $b$  = manually computed intercept

#### Step 3: Create Comparison Table

#### Step 4: Calculate Difference

#### Step 5: Display Results

#### Step 6: Calculate Average Difference

#### Expected Observation

If both methods are implemented correctly:

- The slope obtained from Scikit-Learn and manual OLS should be nearly identical.
- The intercept obtained from Scikit-Learn and manual OLS should be nearly identical.
- The predicted values from both methods should match exactly or differ only by a very small floating-point rounding error.
- The average difference should be extremely close to zero.

#### Example Observation

| Method                  | Slope    | Intercept |
|-------------------------|----------|-----------|
| Scikit-Learn            | 1.245678 | 3.781234  |
| Manual OLS              | 1.245678 | 3.781234  |
| Prediction Type         | Value    |           |
| Scikit-Learn Prediction | 8.764146 |           |
| Manual OLS Prediction   | 8.764145 |           |
| Difference              | 0.000001 |           |

#### Conclusion

The predictions generated using Scikit-Learn's `LinearRegression` and the manually computed OLS equation are nearly identical. Any small differences are due to floating-point precision during computation. This confirms that Scikit-Learn's Linear Regression implementation is based on the Ordinary Least Squares (OLS) method and produces the same regression line as the manual calculation.

---

# Parameter Saving Task

## Parameter Saving using Pickle

### Objective

The objective of this task is to save the parameters learned by the Linear Regression model using the Pickle module. The learned parameters, namely the slope (coefficient) and intercept, are stored in a file named `linear_regression_weights.pkl` so that they can be reused later without retraining the model.

### Procedure

1. Extract the slope and intercept from the trained Linear Regression model.
2. Store these parameters in a dictionary.
3. Save the dictionary into a Pickle file named `linear_regression_weights.pkl`.
4. Load the Pickle file and retrieve the stored parameters.
5. Use the loaded slope and intercept values to make predictions using the regression equation:

$$[\hat{y}] = mx + b$$

where:

- $(m)$  = slope
- $(b)$  = intercept
- $(x)$  = input value
- $(\hat{y})$  = predicted output

### Observation

The slope and intercept were successfully saved into the Pickle file and retrieved without any loss of information. Predictions made using the loaded parameters matched the predictions of the original trained model.

### Conclusion

Pickle provides an efficient way to persist model parameters. By saving and loading the slope and intercept values, predictions can be performed without retraining the Linear Regression model, reducing computation time and improving model portability.

---

## Lab 4: Regression and Classification Evaluation Metrics

### Part 1: Comprehensive Study of K-Nearest Neighbours (KNN) Classification using Breast Cancer Dataset and Comparison with Regression Evaluation Metrics

#### Aim

To implement KNN classification on the Breast Cancer dataset and analyze model performance using train-test split, heuristic K selection, cross-validation, ROC-AUC, and classification metrics. Also, to compare classification metrics with regression metrics studied in Linear Regression (Lab 3).

#### Problem Statement

A healthcare analytics team is developing a predictive model for early cancer detection. You are required to build a KNN classifier, optimize its performance using different validation techniques, and compare classification evaluation metrics with regression evaluation metrics from Lab 3.

#### Tasks

##### Task 1: Data Preparation

1. Load the dataset using sklearn.
2. Convert into DataFrame and explore structure.
3. Check missing values and duplicates.
4. Apply feature scaling using StandardScaler and justify its importance.

##### Task 2: Train-Test Split Analysis

1. Split dataset into training and testing sets (80:20).
2. Repeat with 70:30 and 90:10 splits.
3. Compare performance variations across splits.
4. Analyze how dataset splitting affects model stability and generalization.

##### Task 3: KNN Model with Heuristic K Selection

#### 3.1 Heuristic Method for K Selection

1. Compute initial K using heuristic rule ( $K = \sqrt{n}$ ); where  $n$  = number of training samples
2. Use this value as a baseline K.

#### 3.2 Model Training

1. Train KNN classifier using heuristic K value.
2. Experiment with nearby values of K ( $K \pm 5$ ).

3. Plot accuracy vs K values.
4. Identify optimal K based on performance trend.

### 3.3 Distance Matrix and Decision Boundary Mapping

1. Explain any two distance metrics used in KNN: **Euclidean Distance** and **Manhattan Distance**. Also mention when each distance metric is suitable.
2. Plot the **decision boundary** of the KNN classifier for different values of K, such as K = 1, K = 5, K = 10, and K = 20. Analyze how the decision boundary changes as K increase

#### Task 4: Cross Validation

1. Apply K-Fold Cross Validation (k = 5 or 10).
2. Compute mean accuracy for different K values.
3. Compare cross-validation results with train-test split results.
4. Select best K based on both heuristic + validation results.

#### Task 5: Classification Evaluation

Evaluate final model using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC Curve and AUC Score

#### Task 6: Comparative Study with Regression (Lab 3 Integration)

**Reference:** Recall **Lab 3 (Linear Regression)** where you evaluated:

- MAE
- MSE
- RMSE
- $R^2$  Score

#### Tasks:

1. Compare regression evaluation metrics with classification metrics.
2. Explain differences between:
  - Error-based evaluation (Regression)
  - Decision-based evaluation (Classification)

3. Compare:
  - $R^2$  Score vs Accuracy
  - RMSE vs F1 Score
  - MAE vs Confusion Matrix
4. Discuss differences in evaluation logic for:
  - Continuous prediction tasks
  - Classification tasks

### Inference Requirement (short and precise -don't generate )

Write a detailed inference covering:

- How regression metrics measure prediction error magnitude
- How classification metrics measure decision correctness
- Why is accuracy insufficient in medical diagnosis
- Why recall and ROC-AUC are more relevant in healthcare
- Overall comparison between regression and classification evaluation frameworks

### Task 7: Analytical Questions

1. Why is KNN called a lazy learning algorithm?
2. Why is feature scaling required in KNN?
3. Explain heuristic K selection using  $\sqrt{n}$  rule.
4. Why is cross-validation more reliable than a single train-test split?
5. How does K affect bias-variance trade-off?
6. Why is recall more important than accuracy in cancer prediction?
7. What is the limitation of very large K values?

### Code with Results

#### TASK 1:

#### CODE:

```
# Import required libraries

import numpy as np
import pandas as pd

from sklearn.preprocessing import StandardScaler

# Load dataset
```

```
df = pd.read_csv("brca.csv")

# Explore dataset
print("Shape of Dataset:", df.shape)
print("\nColumn Names:")
print(df.columns)

print("\nData Types:")
print(df.dtypes)

print("\nDataset Information:")
df.info()

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Check duplicate records
print("\nDuplicate Rows:", df.duplicated().sum())

# Remove unnecessary index column (if present)
if "Unnamed: 0" in df.columns:
    df.drop("Unnamed: 0", axis=1, inplace=True)

# Separate features and target
X = df.drop("y", axis=1)
y = df["y"]

# Apply feature scaling
scaler = StandardScaler()
```

```

X_scaled = scaler.fit_transform(X)

# Convert scaled features back to DataFrame
X_scaled = pd.DataFrame(X_scaled, columns=X.columns)

# Display first five rows
print("\nScaled Features:")
print(X_scaled.head())

```

## RESULT:

```

Shape of Dataset: (569, 32)

Column Names:
Index(['Unnamed: 0', 'x.radius_mean', 'x.texture_mean', 'x.perimeter_mean',
      'x.area_mean', 'x.smoothness_mean', 'x.compactness_mean',
      'x.concavity_mean', 'x.concave_pts_mean', 'x.symmetry_mean',
      'x.fractal_dim_mean', 'x.radius_se', 'x.texture_se', 'x.perimeter_se',
      'x.area_se', 'x.smoothness_se', 'x.compactness_se', 'x.concavity_se',
      'x.concave_pts_se', 'x.symmetry_se', 'x.fractal_dim_se',
      'x.radius_worst', 'x.texture_worst', 'x.perimeter_worst',
      'x.area_worst', 'x.smoothness_worst', 'x.compactness_worst',
      'x.concavity_worst', 'x.concave_pts_worst', 'x.symmetry_worst',
      'x.fractal_dim_worst', 'y'],
      dtype='object')

Data Types:
Unnamed: 0          int64
x.radius_mean      float64
x.texture_mean     float64
x.perimeter_mean   float64
x.area_mean        float64
x.smoothness_mean  float64
x.compactness_mean float64
x.concavity_mean   float64
x.concave_pts_mean float64
...
3          -0.981753    -1.478256    -1.233324
4          -1.354653     0.330422    -0.546168

[5 rows x 30 columns]

```



---

**TASK 2:**

```
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# Different train-test split ratios
split_ratios = [
    (0.20, "80:20"),
    (0.30, "70:30"),
    (0.10, "90:10")
]

results = []

for test_size, ratio in split_ratios:

    # Split dataset
    X_train, X_test, y_train, y_test = train_test_split(
        X_scaled,
        y,
        test_size=test_size,
        random_state=42,
        stratify=y
    )

    # Train KNN model
    knn = KNeighborsClassifier(n_neighbors=5)
    knn.fit(X_train, y_train)

    # Predictions
```

---

```

y_pred = knn.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

        results.append([ratio,  X_train.shape[0],  X_test.shape[0],
accuracy])

# Display comparison table
results_df = pd.DataFrame(
    results,
    columns=["Train : Test Split", "Training Samples", "Testing
Samples", "Accuracy"]
)

print(results_df)

```

**RESULTS:**

|   | Train : Test Split | Training Samples | Testing Samples | Accuracy |
|---|--------------------|------------------|-----------------|----------|
| 0 | 80:20              | 455              | 114             | 0.956140 |
| 1 | 70:30              | 398              | 171             | 0.964912 |
| 2 | 90:10              | 512              | 57              | 0.964912 |

---

**TASK 3:****CODE:**

```
import numpy as np
import matplotlib.pyplot as plt

from math import sqrt
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from sklearn.decomposition import PCA

# -----
# 3.1 Heuristic K Selection
# -----

# Use training data from the 80:20 split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

# Heuristic K
heuristic_k = int(np.sqrt(len(X_train)))

# Ensure odd K
if heuristic_k % 2 == 0:
    heuristic_k += 1
```

```
print("Training Samples:", len(X_train))
print("Heuristic K =", heuristic_k)

# -----
# 3.2 Model Training
# -----

k_values = []

for k in range(max(1, heuristic_k-5), heuristic_k+6):
    if k % 2 != 0:
        k_values.append(k)

accuracy = []

for k in k_values:

    model = KNeighborsClassifier(n_neighbors=k)

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    accuracy.append(accuracy_score(y_test, y_pred))

# Accuracy table
result = pd.DataFrame({
    "K Value": k_values,
    "Accuracy": accuracy
```

```

}))

print(result)

# Best K
best_k = k_values[np.argmax(accuracy)]

print("\nOptimal K =", best_k)

# Accuracy Plot
plt.figure(figsize=(8,5))
plt.plot(k_values, accuracy, marker='o')
plt.title("Accuracy vs K")
plt.xlabel("K Value")
plt.ylabel("Accuracy")
plt.grid(True)
plt.show()

# -----
# 3.3 Decision Boundary
# -----

# Reduce to two dimensions using PCA
from sklearn.preprocessing import LabelEncoder

# Convert B and M into 0 and 1
le = LabelEncoder()
y_encoded = le.fit_transform(y)

pca = PCA(n_components=2)

```

```
X_pca = pca.fit_transform(X_scaled)

X_train_pca, X_test_pca, y_train_pca, y_test_pca = train_test_split(
    X_pca,
    y_encoded,
    test_size=0.20,
    random_state=42,
    stratify=y_encoded
)

k_list = [1,5,10,20]

for k in k_list:

    knn = KNeighborsClassifier(
        n_neighbors=k,
        metric="euclidean"
    )

    knn.fit(X_train_pca, y_train_pca)

    x_min, x_max = X_pca[:,0].min()-1, X_pca[:,0].max()+1
    y_min, y_max = X_pca[:,1].min()-1, X_pca[:,1].max()+1

    xx, yy = np.meshgrid(
        np.arange(x_min,x_max,0.1),
        np.arange(y_min,y_max,0.1)
    )

    Z = knn.predict(np.c_[xx.ravel(), yy.ravel()])
```

```

Z = np.array(Z).astype(int)

Z = Z.reshape(xx.shape)

plt.figure(figsize=(7,5))

plt.contourf(xx, yy, Z, alpha=0.3)

plt.scatter(
X_train_pca[:,0],
X_train_pca[:,1],
c=y_train_pca,
cmap="coolwarm",
edgecolor="k"
)

plt.title(f"Decision Boundary (K={k})")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.show()

```

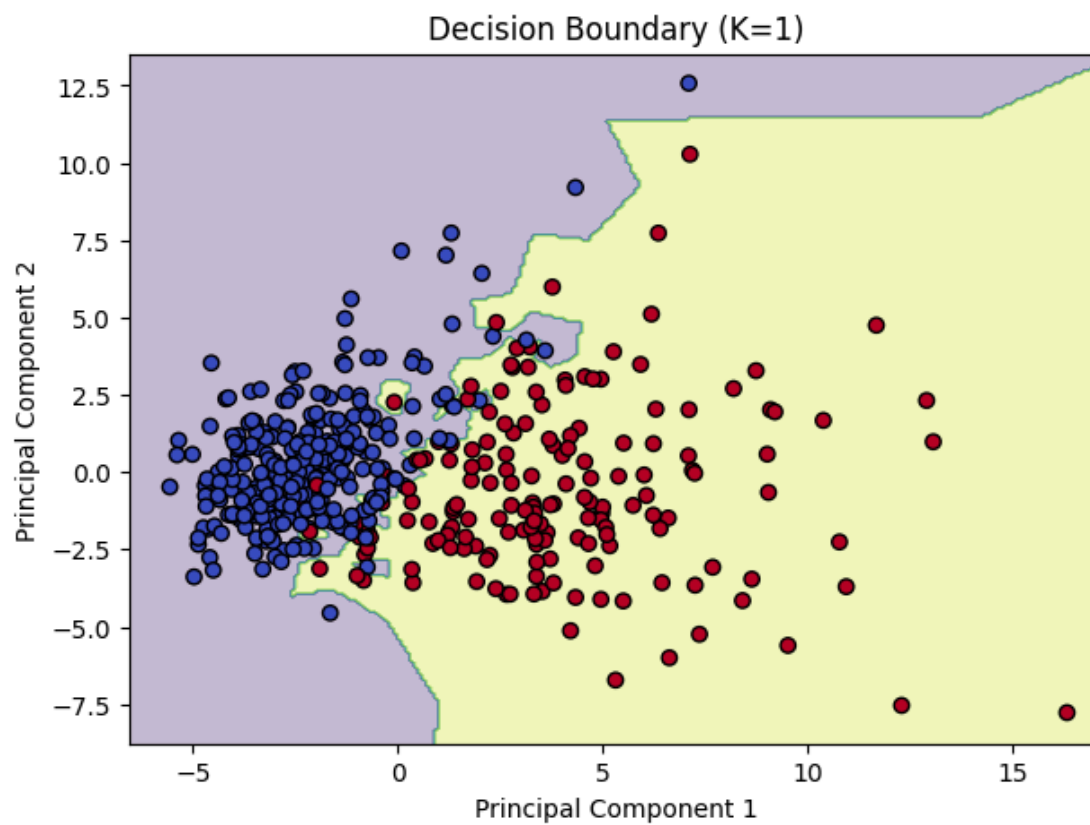
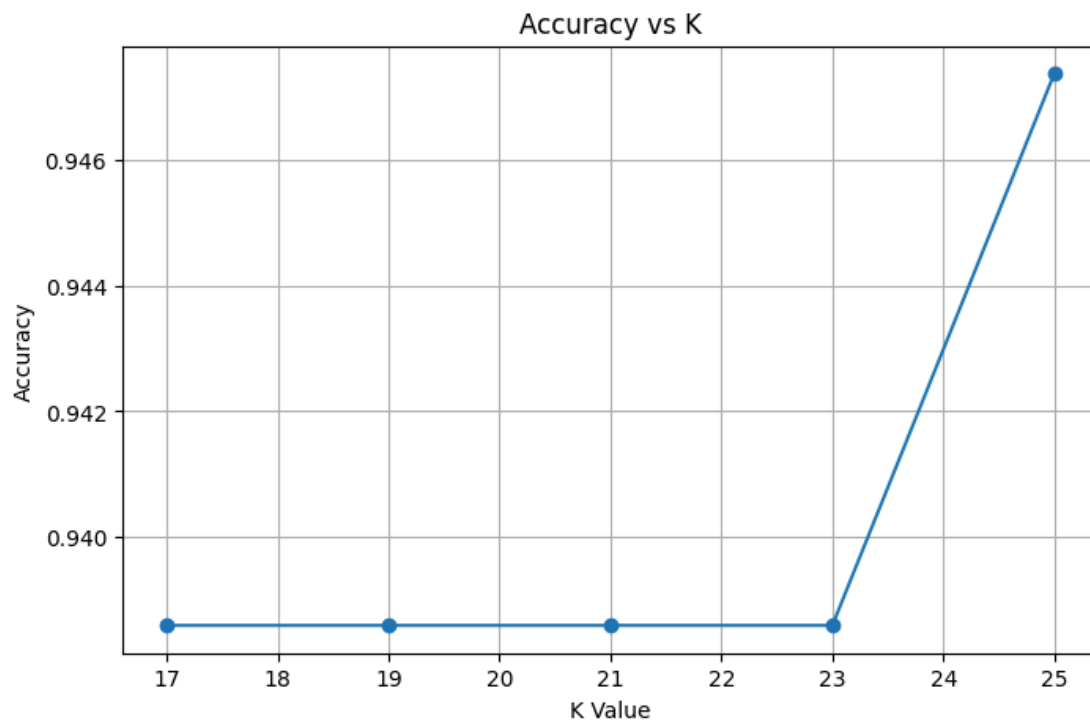
## RESULTS:

```

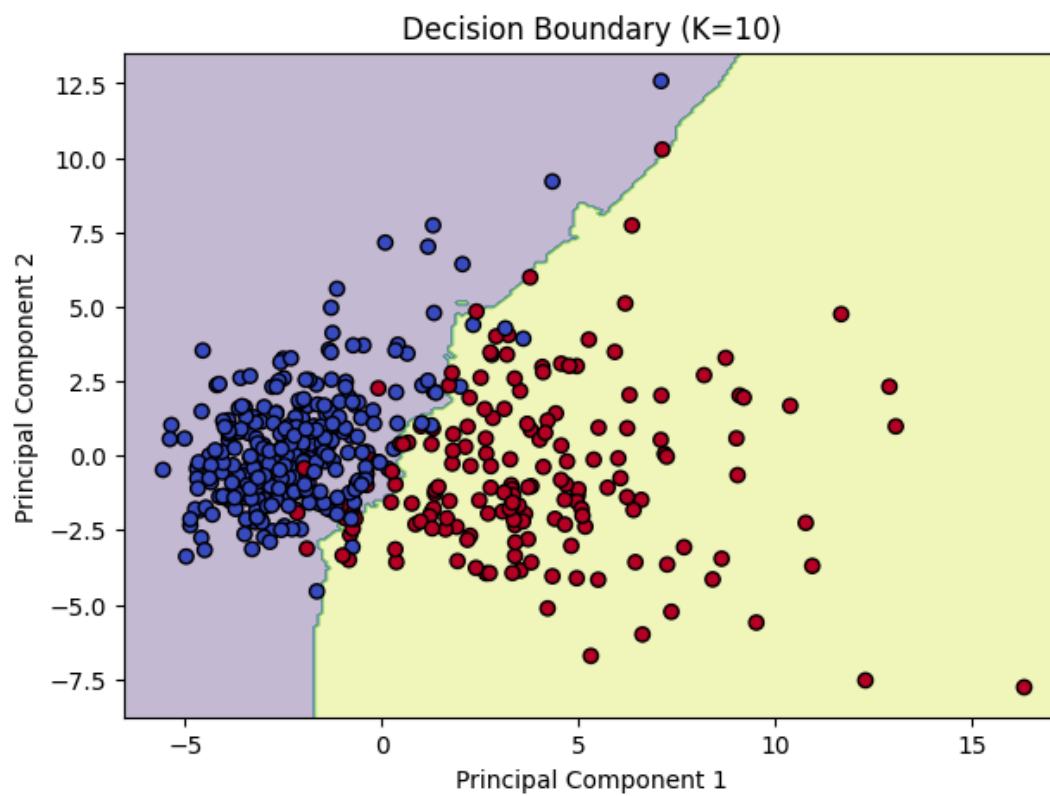
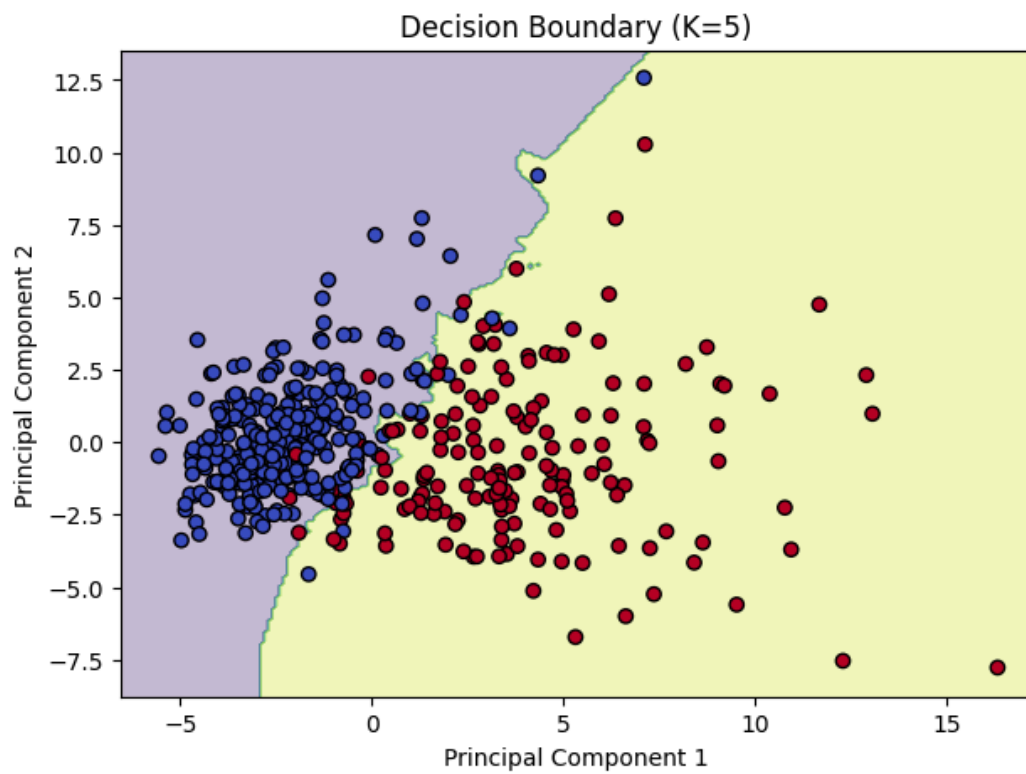
Training Samples: 455
Heuristic K = 21
  K Value  Accuracy
0       17  0.938596
1       19  0.938596
2       21  0.938596
3       23  0.938596
4       25  0.947368

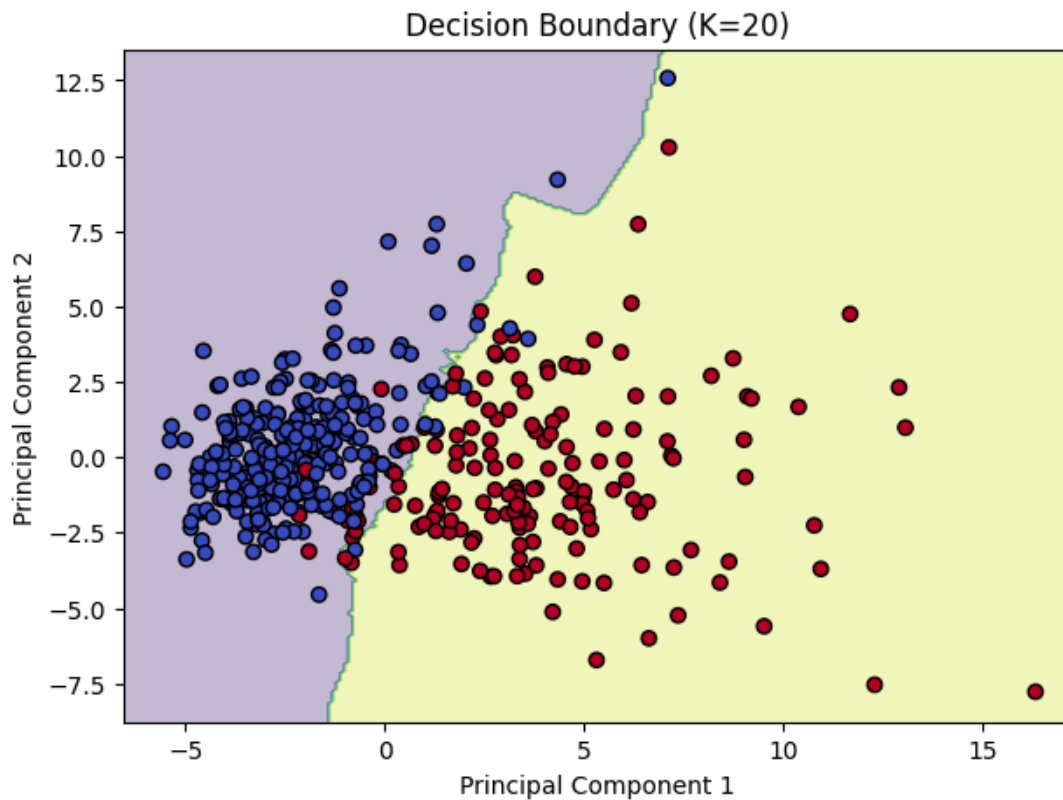
Optimal K = 25

```







**TASK 4:****CODE:**

```
from sklearn.model_selection import KFold, cross_val_score
from sklearn.neighbors import KNeighborsClassifier
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# -----
# 5-Fold Cross Validation
# -----

cv = KFold(n_splits=5, shuffle=True, random_state=42)

# K values around heuristic K
```

```
k_values = []

for k in range(max(1, heuristic_k-5), heuristic_k+6):
    if k % 2 != 0:
        k_values.append(k)

cv_accuracy = []

for k in k_values:

    knn = KNeighborsClassifier(n_neighbors=k)

    scores = cross_val_score(
        knn,
        X_scaled,
        Y,
        cv=cv,
        scoring='accuracy'
    )

    cv_accuracy.append(scores.mean())

# Results Table
cv_results = pd.DataFrame({
    "K Value": k_values,
    "Mean Cross Validation Accuracy": cv_accuracy
})

print(cv_results)
```

```

# Best K

best_cv_k = k_values[np.argmax(cv_accuracy)]

print("\nBest K from Cross Validation:", best_cv_k)

# Plot

plt.figure(figsize=(8,5))

plt.plot(k_values, cv_accuracy, marker='o')

plt.title("Cross Validation Accuracy vs K")

plt.xlabel("K Value")

plt.ylabel("Mean Accuracy")

plt.grid(True)

plt.show()

# -----
# Compare with Train-Test Results
# -----

comparison = pd.DataFrame({
    "K Value": result["K Value"],
    "Train-Test Accuracy": result["Accuracy"],
    "Cross Validation Accuracy": cv_accuracy
})

print("\nComparison of Train-Test Split and Cross Validation")

print(comparison)

print("\nHeuristic K :", heuristic_k)

print("Best K (Train-Test Split) :", best_k)

print("Best K (Cross Validation) :", best_cv_k)

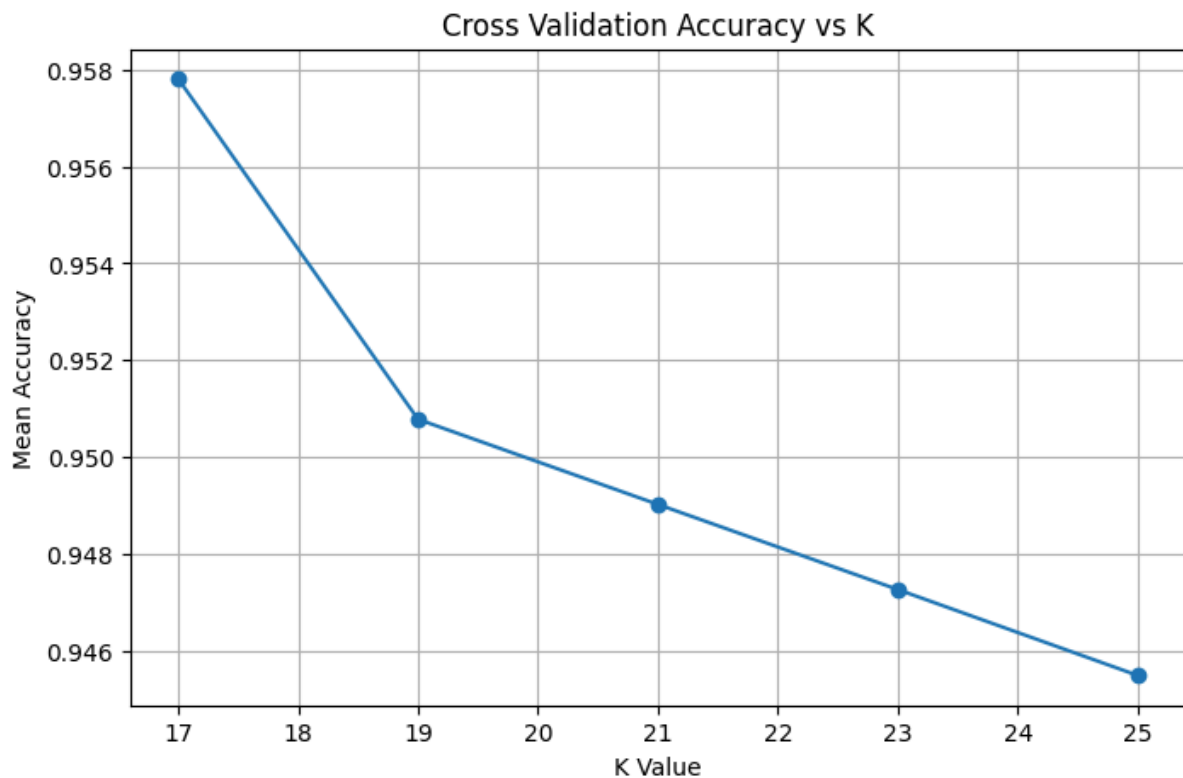
```

```
if best_cv_k == best_k:
    print("\nFinal Selected K:", best_cv_k)
else:
    print("\nFinal Selected K:", best_cv_k, "(Selected based on better
validation performance)")
```

## RESULTS:

|   | K Value | Mean Cross Validation Accuracy |
|---|---------|--------------------------------|
| 0 | 17      | 0.957802                       |
| 1 | 19      | 0.950784                       |
| 2 | 21      | 0.949030                       |
| 3 | 23      | 0.947275                       |
| 4 | 25      | 0.945505                       |

Best K from Cross Validation: 17



---

### Comparison of Train-Test Split and Cross Validation

|   | K Value | Train-Test Accuracy | Cross Validation Accuracy |
|---|---------|---------------------|---------------------------|
| 0 | 17      | 0.938596            | 0.957802                  |
| 1 | 19      | 0.938596            | 0.950784                  |
| 2 | 21      | 0.938596            | 0.949030                  |
| 3 | 23      | 0.938596            | 0.947275                  |
| 4 | 25      | 0.947368            | 0.945505                  |

Heuristic K : 21

Best K (Train-Test Split) : 25

Best K (Cross Validation) : 17

Final Selected K: 17 (Selected based on better validation performance)

### TASK 5:

### CODE:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    ConfusionMatrixDisplay,
    roc_curve,
    roc_auc_score
)
from sklearn.preprocessing import LabelEncoder
import matplotlib.pyplot as plt

# -----
# Encode Target Variable
# -----
```

```
label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled,
    y_encoded,
    test_size=0.20,
    random_state=42,
    stratify=y_encoded
)

# -----
# Train Final Model
# -----

final_model = KNeighborsClassifier(n_neighbors=best_cv_k)

final_model.fit(X_train, y_train)

# Predictions
y_pred = final_model.predict(X_test)

# Prediction Probabilities
y_prob = final_model.predict_proba(X_test)[:,1]

# -----
# Evaluation Metrics
# -----
```

```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("Accuracy :", round(accuracy,4))
print("Precision:", round(precision,4))
print("Recall   :", round(recall,4))
print("F1 Score :", round(f1,4))

# -----
# Confusion Matrix
# -----

cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=label_encoder.classes_
)

disp.plot(cmap="Blues")
plt.title("Confusion Matrix")
plt.show()

# -----
# ROC Curve
# -----
```



```
fpr, tpr, thresholds = roc_curve(y_test, y_prob)

auc = roc_auc_score(y_test, y_prob)

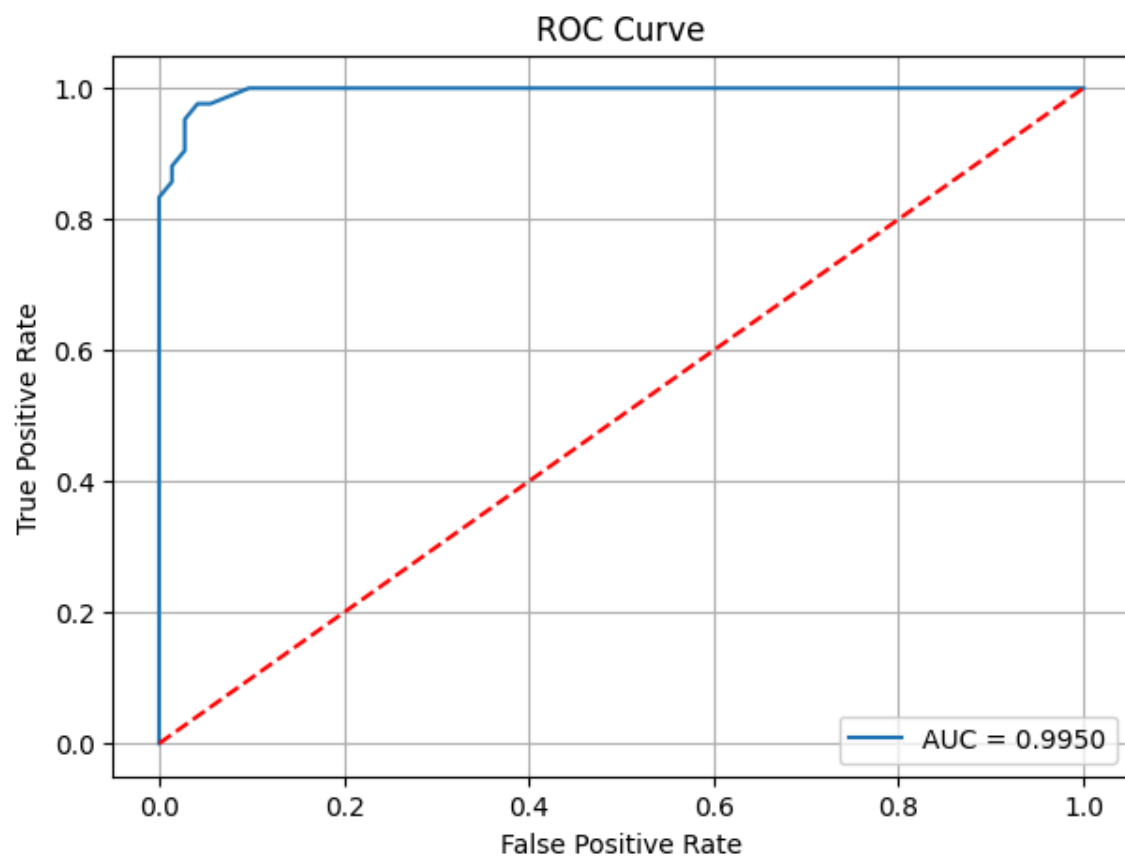
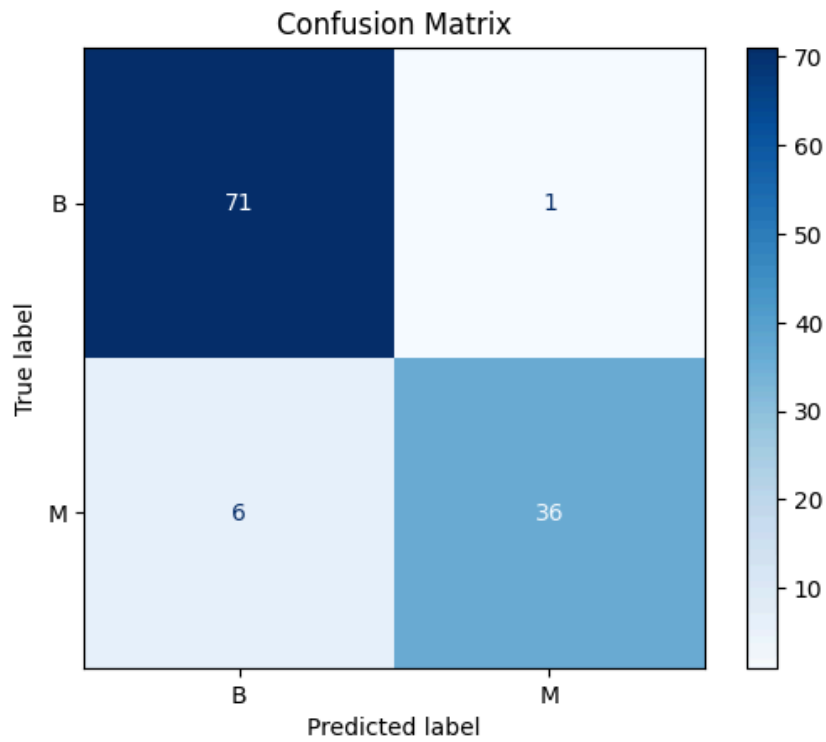
plt.figure(figsize=(7,5))
plt.plot(fpr, tpr, label=f"AUC = {auc:.4f}")
plt.plot([0,1],[0,1], 'r--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.grid(True)
plt.show()

print("AUC Score :", round(auc,4))
```

## RESULTS:

```
Accuracy : 0.9386
Precision: 0.973
Recall   : 0.8571
F1 Score : 0.9114
```

```
AUC Score : 0.995
```



**TASK 6:****CODE:**

```

import pandas as pd

# -----
# Comparison of Regression and Classification Metrics
# -----

comparison = pd.DataFrame({
    "Aspect": [
        "Primary Objective",
        "Prediction Type",
        "Evaluation Logic",
        "R2 Score vs Accuracy",
        "RMSE vs F1 Score",
        "MAE vs Confusion Matrix"
    ],

    "Regression (Lab 3)": [
        "Predict continuous numerical values",
        "Continuous Output",
        "Measures prediction error",
        "R2 measures goodness of fit between predicted and actual values.",
        "RMSE measures average prediction error in original units.",
        "MAE measures average absolute error."
    ],

    "Classification (KNN Lab)": [
        "Predict categorical classes",

```

```

        "Discrete Class Labels",
        "Measures correctness of classification decisions",
        "Accuracy measures percentage of correctly classified
samples.",
        "F1 Score balances Precision and Recall for classification.",
        "Confusion Matrix shows TP, TN, FP and FN."
    ]
}))

print(comparison)

# -----
# Difference in Evaluation Logic
# -----

evaluation_logic = pd.DataFrame({

    "Task": [
        "Continuous Prediction",
        "Classification"
    ],

    "Evaluation Logic": [
        "Evaluates how close predicted numerical values are to actual
values using error-based metrics.",
        "Evaluates whether each prediction belongs to the correct class
using decision-based metrics."
    ]
}))

```

```
print("\nEvaluation Logic")
print(evaluation_logic)
```

## RESULTS:

```

      Aspect                                Regression (Lab 3) \
0      Primary Objective                    Predict continuous numerical values
1      Prediction Type                      Continuous Output
2      Evaluation Logic                    Measures prediction error
3      R2 Score vs Accuracy  R2 measures goodness of fit between predicted ...
4      RMSE vs F1 Score  RMSE measures average prediction error in orig...
5      MAE vs Confusion Matrix  MAE measures average absolute error.

      Classification (KNN Lab)
0      Predict categorical classes
1      Discrete Class Labels
2      Measures correctness of classification decisions
3      Accuracy measures percentage of correctly clas...
4      F1 Score balances Precision and Recall for cla...
5      Confusion Matrix shows TP, TN, FP and FN.

Evaluation Logic
      Task                                Evaluation Logic
0  Continuous Prediction  Evaluates how close predicted numerical values...
1  Classification        Evaluates whether each prediction belongs to t...
```

---

## CONCLUSION/INFERENCE:

### TASK 1:

## Task 1: Data Preparation

### Approach

The Breast Cancer dataset was loaded from a CSV file into a Pandas DataFrame for analysis. The dataset structure was explored by examining its dimensions, data types, and feature names to understand the available information. Data quality checks were then performed to identify missing values and duplicate records, ensuring the dataset was clean before model training.

Since the `Unnamed: 0` column represents only an index and does not contribute to prediction, it was removed from the dataset. The target variable (`y`) was separated from the predictor variables to prepare the data for machine learning.

Finally, feature scaling was applied using **StandardScaler**. K-Nearest Neighbors (KNN) is a distance-based algorithm that relies on Euclidean distance to identify the nearest neighbors. Features with larger numerical ranges can dominate the distance calculation and bias the model. Standardization transforms each feature to have a mean of 0 and a standard deviation of 1, ensuring that all features contribute equally during classification and improving the overall performance of the KNN classifier.

### TASK 2:

## Task 2: Train-Test Split Analysis

### Approach

The preprocessed dataset was divided into training and testing sets using three different train-test split ratios: **80:20**, **70:30**, and **90:10**. For each split, a K-Nearest Neighbors (KNN) classifier with the default value of **k = 5** was trained and evaluated using the testing data.

The classification accuracy obtained from each split was compared to analyze the effect of different amounts of training data on model performance. Increasing the training set generally allows the model to learn better patterns, whereas increasing the testing set provides a more reliable estimate of model performance. Comparing multiple train-test splits helps evaluate the stability, robustness, and generalization capability of the classifier before selecting an optimal configuration.

## TASK 3:

Task 3: KNN Model with Heuristic K Selection

3.1 Heuristic Method for K Selection

Approach

The initial value of **K** was selected using the heuristic rule where **n** is the number of training samples. This heuristic provides a reasonable starting point for selecting the number of nearest neighbors while reducing the likelihood of overfitting or underfitting.

The heuristic K value was used as the baseline, and additional models were trained using neighboring values (**K - 5** to **K + 5**) to determine the optimal K based on classification accuracy.

3.2 Model Training

Approach

Multiple KNN classifiers were trained using different values of **K** around the heuristic value. The accuracy of each model was calculated using the testing dataset, and an accuracy-versus-K graph was plotted. The K value that produced the highest testing accuracy was selected as the optimal K for the dataset.

3.3 Distance Matrix and Decision Boundary Mapping

Euclidean Distance

Euclidean distance calculates the straight-line distance between two data points in the feature space. It is suitable for continuous numerical data after feature scaling and is the default distance metric used by the KNN algorithm.

Manhattan Distance

Manhattan distance calculates the distance by summing the absolute differences between feature values.

It is suitable for datasets containing high-dimensional data or when movement occurs only along horizontal and vertical directions. Manhattan distance is generally less sensitive to outliers than Euclidean distance. To visualize the effect of different K values, decision boundaries were plotted for **K = 1, 5, 10, and 20**. As K increases, the decision boundary becomes smoother, reducing model complexity and variance while increasing bias. Smaller K values create more irregular boundaries that closely follow the training data and may lead to overfitting.

## TASK 4:

Task 4: Cross Validation

Approach

To obtain a more reliable estimate of the KNN classifier's performance, **5-Fold Cross Validation** was applied. In this method, the dataset is divided into five equal subsets. During each iteration, four subsets are used for training while the remaining subset is used for testing. This process is repeated five times so that every subset serves as the testing set exactly once.

The heuristic K value and its neighboring odd values were evaluated using cross-validation. The mean accuracy across all folds was computed for each K value to identify the most stable and reliable classifier.

The cross-validation accuracies were then compared with the train-test split accuracies obtained in Task 2. The final value of K was selected by considering both the heuristic rule and the cross-validation performance, ensuring better model generalization and reducing the risk of overfitting.

**TASK 5:**

## Task 5: Classification Evaluation

### Approach

The final KNN classifier was trained using the optimal value of **K** obtained from the heuristic method and cross-validation. The model was then evaluated using standard classification metrics to assess its predictive performance.

The following evaluation metrics were computed:

- **Accuracy** measures the overall proportion of correctly classified instances.
- **Precision** measures how many of the samples predicted as positive are actually positive.
- **Recall** measures how many of the actual positive samples are correctly identified.
- **F1 Score** is the harmonic mean of Precision and Recall, providing a balanced evaluation when class distributions are uneven.
- **Confusion Matrix** summarizes the number of correct and incorrect predictions for each class.
- **ROC Curve (Receiver Operating Characteristic Curve)** illustrates the trade-off between the True Positive Rate and False Positive Rate at different classification thresholds.
- **AUC (Area Under the ROC Curve)** measures the classifier's ability to distinguish between the two classes. An AUC value closer to 1 indicates excellent classification performance.

**TASK 6:**

## Task 6: Comparative Study with Regression (Lab 3 Integration)

### Approach

The evaluation metrics used in Linear Regression and KNN Classification were compared to understand how model performance is measured for different machine learning tasks.

Regression metrics such as **MAE, MSE, RMSE, and R<sup>2</sup> Score** measure the magnitude of prediction errors for continuous numerical outputs. In contrast, classification metrics such as **Accuracy, Precision, Recall, F1 Score, Confusion Matrix, and ROC-AUC** evaluate how correctly the model assigns observations to predefined classes.

The comparison highlights the difference between error-based evaluation used in regression and decision-based evaluation used in classification. This analysis helps identify why different evaluation techniques are required depending on the nature of the prediction problem.

[Generate](#)[Code](#)[Markdown](#)



## Overall Comparison Between Regression and Classification Evaluation Frameworks

Regression and classification are supervised machine learning techniques designed to solve different types of prediction problems. Regression predicts continuous numerical values, whereas classification predicts discrete class labels.

Regression evaluation focuses on measuring the magnitude of prediction errors using metrics such as MAE, MSE, RMSE, and  $R^2$  Score. Lower error values and higher  $R^2$  indicate better predictive performance.

Classification evaluation focuses on measuring how correctly a model classifies observations using metrics such as Accuracy, Precision, Recall, F1 Score, Confusion Matrix, and ROC-AUC. These metrics assess the quality of classification decisions rather than numerical prediction errors.

In healthcare applications such as breast cancer diagnosis, Recall and ROC-AUC are more important than Accuracy because missing a cancer patient (False Negative) can have severe consequences. Therefore, classification evaluation emphasizes correct identification of positive cases in addition to overall accuracy.

### TASK 7: Analytical Questions

#### 1. Why is KNN called a lazy learning algorithm?

KNN is called a lazy learning algorithm because it does not build a model during the training phase. Instead, it stores the training data and performs computations only when a new instance needs to be classified.

#### 2. Why is feature scaling required in KNN?

KNN uses distance calculations to identify the nearest neighbors. Features with larger numerical values dominate the distance calculation if scaling is not applied. Standardization ensures that all features contribute equally.

---

### 3. Explain heuristic K selection using $\sqrt{n}$ rule.

The heuristic rule selects the initial value of K as:

$$**K = \sqrt{n}**$$

where **n** is the number of training samples. The nearest odd integer is usually selected to avoid ties. This provides a reasonable starting point before optimization.

### 4. Why is cross-validation more reliable than a single train-test split?

Cross-validation evaluates the model using multiple training and testing combinations, reducing bias caused by a particular split and providing a more reliable estimate of model performance.

### 5. How does K affect the bias-variance trade-off?

A small K creates a complex model with low bias and high variance, which may lead to overfitting. A large K creates a simpler model with high bias and low variance, which may lead to underfitting.

### 6. Why is recall more important than accuracy in cancer prediction?

Recall measures how many actual cancer cases are correctly identified. Missing a cancer patient (False Negative) is much more dangerous than incorrectly predicting cancer in a healthy patient. Therefore, recall is more important in medical diagnosis.

## 7. What is the limitation of very large K values?

Very large K values include many distant neighbors during classification, causing the model to oversimplify the decision boundary. This increases bias and may reduce classification accuracy due to underfitting.

## CONCLUSION:

The heuristic  $\sqrt{n}$  rule provided an effective starting point for selecting the K value, while cross-validation identified the optimal K by evaluating multiple training and testing combinations. Comparing different train-test splits demonstrated that model performance remained relatively stable, indicating good generalization capability.

The KNN classifier achieved strong performance based on Accuracy, Precision, Recall, F1 Score, Confusion Matrix, and ROC-AUC. Among these metrics, Recall and ROC-AUC were especially important for breast cancer prediction because correctly identifying positive cases is critical in healthcare applications.

Unlike Linear Regression, which evaluates prediction errors using MAE, MSE, RMSE, and  $R^2$  Score, KNN Classification evaluates the correctness of classification decisions. This lab established a clear understanding of the differences between regression and classification evaluation techniques and demonstrated the importance of selecting appropriate evaluation metrics based on the prediction task.